Maths for Economists

A Comprehensive Handbook

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7th March 2026

Preface

This book is written as a handbook for students of economics rather than a course note for a specific subject like MathEcon. To make it a comprehensive one, I'll try to cover almost all maths needed for studying economics, including logic, algebra, analysis, topology, probability theory and optimization theory. Furthermore, I will incorporate additional topics, drawing primarily from the syllabi of courses I have completed at SUFE. In this way, those (especially from SUFE) who are taking courses like Mathematical Analysis can refer to it as a supplementary resource. However, considering the vast range of knowledge required, it's NOT possible to cover everything, thus the book is written with an emphasis on the mathematical foundations for economics.

Though serving as a handbook, the book will be written as intuitively as possible. I believe this book is suitable for self-taught learners as we're biulding all the knowledge step by step while reading it. This approach, I believe, mirrors the way many economists think: by providing an intuitive understanding of how different branches of knowledge are developed and structured as a whole, the definitions and techniques can feel natural and logical to the reader.

I recommend treating this book, which is written in a modern approach, as a second course after you complete your first year of study. A solid grasp of the basics of linear algebra and mathematical analysis is required as prerequisites so that the modern approach is understandable. The book is under construction and may be updated on a monthly basis.

Acknowledgement

This book is written using LaTeX. The template is based on Agni Datta's "Preprint Book Manuscript Template", which I have modified to fit my requirements. The famous ElegantLaTeX template has also inspired me greatly. The source code is available in my GitHub repository for reference.

I would like to thank Copilot and DeepSeek for helping me write and edit this book.

Contents

I Foundations **1**

1 Logic **3**

1.1 Negation . . . 3

1.2 Conjunction and Disjunction . . . 4

1.3 Quantifiers . . . 4

1.3.1 Property . . . 4

1.3.2 Forall and Exists . . . 4

1.4 Implications . . . 5

1.4.1 Implication . . . 5

1.4.2 Equivalence . . . 5

1.4.3 Prove by Contrapositive . . . 5

1.4.4 Prove by Contradiction . . . 6

2 Set, Relation and Operation **7**

2.1 Operations of Sets . . . 7

2.1.1 Complement, Intersection and Union . . . 7

2.1.2 Power Set . . . 8

2.1.3 Cartesian Product . . . 8

2.1.4 Family of Sets . . . 9

2.2 Relation and Operation . . . 10

2.2.1 Equivalence . . . 11

2.2.2 Order . . . 12

2.2.3 Mapping . . . 14

2.2.4 Correspondence 15
2.2.5 Operation 15
2.3 Countability 16
2.3.1 The Axiom of Choice 16
2.3.2 Cardinality 16
2.3.3 Countable Set 17

3 Group, Ring and Field **21**
3.1 Group 21
3.1.1 the Definition 21
3.1.2 Homomorphisms 22
3.1.3 Isomorphisms 23
3.2 Ring and Field 24
3.2.1 The Definition 24
3.2.2 Homomorphism and Isomorphism 24
3.2.3 Ordered Field 25
3.3 Polynomials 25

II Vector Space and Linear Algebra **27**

4 Vector Space **29**
4.1 the Definition of Vector Space and Subspaces 29
4.2 Operation of Subspaces 30
4.2.1 Sum 30
4.2.2 Direct Sum 30
4.3 Subspace Spanned by Vectors 31
4.3.1 Span 31
4.3.2 Linear Independence 31
4.4 Finite Dimensional Vector Space 33
4.4.1 Basis 33
4.4.2 Dimension 34

5 Linear Mapping and Matrix **35**
5.1 Linear Mapping 35
5.1.1 the Definitions 35

5.1.2 Isomorphic Linear Mapping . . . 36
5.2 Matrix . . . 37
5.2.1 From Vector to Coordinate . . . 39
5.2.2 From Linear Mapping to Matrix . . . 39
5.2.3 Operations of Matrix . . . 40
5.2.4 Change of Basis . . . 43

6 Normed Vector Space 45

7 Multilinear Form and Determinant 47
7.1 Bilinear Form . . . 49
7.1.1 Symmetric and Alternating Bilinear Form . . . 50
7.1.2 Matrix of a Bilinear Form . . . 51
7.1.3 Quadratic Form . . . 53
7.2 Multilinear Form . . . 53
7.2.1 Alternating Multilinear Form . . . 53
7.2.2 Swapping the Entries . . . 54
7.3 Determinant . . . 55
7.3.1 From Alternating Multilinear Form to Determinant . . . 55
7.3.2 Calculation of Determinants . . . 57
7.3.3 Property of Determinants . . . 57

8 Eigenvalue and Eigenvector 59

III Metric Space and Topology Space 61

9 Metric Space and Convergence 63
9.1 Metric Space . . . 63
9.1.1 the Definitions . . . 63
9.1.1.1 Diameter . . . 64
9.1.1.2 Neighborhood . . . 64
9.1.1.3 Cluster Point . . . 65
9.1.2 Induced Metric . . . 65
9.2 Convergence . . . 65
9.2.1 Convergence of a Sequence . . . 65

9.3 Improper Convergence 67
9.3.1 Infinite Limit 67
9.3.2 Some Special Sequences 68
9.3.2.1 Monotone Sequence 68
9.3.2.2 Subsequence 68
9.3.3 Limit Superior and Limit Inferior 68
9.4 Completeness 69
9.4.1 Cauchy Sequence 69
9.4.2 Banach Space 69

10 Topology Space **71**
10.1 the Definition 71
10.1.1 Open Set and Closed Set 72
10.1.2 Interior, Closure and Boundary 73
10.1.3 Topological Basis 75
10.2 Continuous Function 77
10.2.1 Continuity in Metric Space 77
10.2.2 Continuity 78
10.2.3 Homomorphism 79
10.3 Compactness 79
10.3.1 the Basics 79
10.3.2 Compactness of Product Space 79
10.3.3 Compactness of Induced Topology Space 79
10.4 Connectedness 79
10.5 Constructing Topology 79
10.5.1 Topology Subspace 79
10.5.2 Product Space 80
10.5.3 Quotient Space 80
10.5.4 Topology Space Induced by Metric 80

IV Differentiation in One Variable **81**

V More in Convergence: Uniform Convergence **83**

A Notation **85**

B Supplementary Scripts **87**

Part I

Foundations

Chapter 1

Logic

To make complex ideas clear, we often separate them into statements and check whether they're right or wrong. Here we use symbolic logic. Symbolic logic is basically about statements which can be claimed to be either true or false and "operations" of statements.

We start with what is a statement. For sure, most of the sentences can be seen as a statement. But to notice, not all sentences are statements. For example, "This sentence is false." is not a statement, that is because when it is true, it is false and vice versa.

In maths, we call statements as propositions and we can categorize them into theorems, axioms, lemmas, corollaries, etc. Axioms are statements that we accept to be true without proof. Theorems are statements that can be proven to be true based on axioms and previously proven theorems. Lemmas are "helper" theorems used to prove larger theorems and corollaries are statements that follow directly from a theorem.

1.1 Negation

We use the symbol $\neg$ to express "NOT". Suppose A is a statement, then $\neg A$ is also a statement and $\neg A$ means "A is NOT true". We can use a truth table to show the

relationship between A and $\neg A$.

Statement	A	$\neg A$
Truth	T	F
Value	F	T

1.2 Conjunction and Disjunction

Conjunction and Disjunction are ways to combine statements together. We use the symbol $\wedge$ to express "AND". Suppose A and B are statements, then $A \wedge B$ is also a statement and $A \wedge B$ is true only when both A and B are true. Similarly, we use the symbol $\vee$ to express "OR". Suppose A and B are statements, then $A \vee B$ is also a statement and $A \vee B$ is false only when both A and B are false.

We can use a truth table to show the relationship between A, B, $A \wedge B$ and $A \vee B$.

Statement	A	B	$A \wedge B$	$A \vee B$
	T	T	T	T
Truth	T	F	F	T
Value	F	T	F	T
	F	F	F	F

1.3 Quantifiers

1.3.1 Property

We first define the concept of property. We say $P(x)$ is a property of x (x is from a particular class) if when x is replaced with a certain object, $P(x)$ becomes a statement. For example, let $P(x)$ be "x is even", then $P(2)$ is true and $P(3)$ is false. The set $\{x : P(x)\}$ consists of all values of x such that $P(x)$ is true.

1.3.2 Forall and Exists

Now we can create another kind of statement using quantifiers. There are two quantifiers: "for all" and "there exists". The expression $\forall x \in X : P(x)$ means "for all

components x in the set X, $P(x)$ is true". And the expression $\exists x \in X : P(x)$ means "there exists at least one component x in the set X, such that $P(x)$ is true".

A very important proposition is that

$$\neg[\exists x \in X : P(x)] = \forall x \in X : \neg P(x) \tag{1.1}$$

And to understand all statements, we also need $\neg(\neg A) = A, \neg(A \wedge B) = \neg A \vee \neg B, \neg(A \vee B) = \neg A \wedge \neg B$. This becomes trivial to deal with quantifiers that we only have to interchange $\vee$ and $\wedge$ and change $\forall$ to $\exists$ and vice versa when we negate a statement.

1.4 Implications

1.4.1 Implication

The implication $(A \implies B) := (\neg A) \vee B$ is false if and only if A is true and B is false. The definition is simple, but note that $A \implies B$ not necessarily mean that A causes B to be true. When A and B are false, the implication is also true, which makes it different from the familiar meaning of "imply".

We also say that A is a sufficient condition for B, and B is a necessary condition for A when we write the statement $A \implies B$.

1.4.2 Equivalence

When A is both necessary and sufficient for B, we say that A is equivalent to B, and we write $A \iff B$. Note that $A \iff B$ is equivalent to $(A \implies B) \wedge (B \implies A)$.

1.4.3 Prove by Contrapositive

With equivalence, we know that the equation in (1.1) actually means equivalent and we can make clear of the above relationships. Moreover, using the interchanging technique, we can find the counterpositive statement

$$(A \implies B) \iff (\neg B \implies \neg A)$$

This inspires us to prove a statement, we can turn to its contrapositive if difficulties arise when proving the original statement.

1.4.4 Prove by Contradiction

Consider statements $(A \implies C), (C \implies B)$, then

$$(A \implies B) \iff (A \implies C) \wedge (C \implies B)$$

Suppose B is false and assume A is true, then $(C \implies B)$ and $(A \implies C)$ are false, that is to say, we can find a statement C with the false truth value.

This leads us to the proof by contradiction method. To prove $A \implies B$, we can assume A is true and B is false, then try to find a statement with a false truth value and we prove the original statement.

Chapter 2

Set, Relation and Operation

Since all readers should be familiar with the high-school-level set theory, we only give a brief review of the basic concepts and operations of sets.

We can't define what is a set but we can give it a description. A set is all objects that satisfy a certain property. The following lists some of the basics.

Definition 2.0.1 (Subset)**.** Let A, B be sets, when we say $A \subset B$, we mean $\forall x \in A, x \in B$. Notice

$$A = B \iff (A \subset B) \wedge (B \subset A)$$

2.1 Operations of Sets

2.1.1 Complement, Intersection and Union

Definition 2.1.1 (Complement, Intersection and Union)**.** Let A, B be sets, we define the following operations:

Complement The (relative) complement of set B in set A is defined as $A \setminus B = \{x : (x \in A) \wedge (x \notin B)\}$.

Intersection The intersection of sets A and B is defined as $A \cap B = \{x : (x \in A) \wedge (x \in B)\}$.

Union The union of sets A and B is defined as $A \cup B = \{x : (x \in A) \vee (x \in B)\}$.

By definition, one can easily check:

Proposition 2.1.1. *Let A, B, C be sets, then*

Commutativity $A \cap B = B \cap A$ *and* $A \cup B = B \cup A$.

Associativity $(A \cap B) \cap C = A \cap (B \cap C)$ *and* $(A \cup B) \cup C = A \cup (B \cup C)$.

Distributivity $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ *and* $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

de Morgan's Laws $(A \cap B)^c = A^c \cup B^c$ *and* $(A \cup B)^c = A^c \cap B^c$

Subset $A \subset B \iff A \cap B = A \iff A \cup B = B$.

2.1.2 Power Set

Definition 2.1.2 (Power Set)**.** Let A be a set, define:

$$2^A = \{X : X \subset A\}$$

2.1.3 Cartesian Product

We define an ordered pair or a n-tuple $x = (x_1, x_2, \cdots, x_n)$ and the equity means that all components are equal. Denote $x_j := pr_j(x)$ and we call it the jth projection of x.

The Cartesian product is to describe the set of ordered pairs.

Definition 2.1.3 (Cartesian Product)**.** Let A, B be sets, define:

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

and by deduction,

$$\prod_{i=1}^{n} A_i = (A_1 \times \cdots \times A_{n-1}) \times A_n$$

Specially, when $A_i = A$ for all i, we denote:

$$A^n = \prod_{i=1}^{n} A$$

To practice logic, we show how to prove the following proposition.

Proposition 2.1.2. *Let A, B be sets, then $A \times B = \emptyset \iff (A = \emptyset) \vee (B = \emptyset)$*

Proof. ($\Rightarrow$) We prove by contradiction. Suppose $A \times B$ is non-empty, which implies there exists $(a, b) \in A \times B$, then by definition of Cartesian product, we have $a \in A$ and $b \in B$, contradicting the assumption.

($\Leftarrow$) This side can be similarly proved by contradiction. □

2.1.4 Family of Sets

We now extend the operation of intersection and union a little bit. Consider we're intersecting a series of sets $A, B, C, \cdots$, we first rename the sets as $\{A_i\}_{i \in I}$, where I is called an index set, and intersect them according to the index i, that is

$$\bigcap_{i \in I} A_i = \{x : \forall i \in I, x \in A_i\}.$$

Similarly, for union, we have

$$\bigcup_{i \in I} A_i = \{x : \exists i \in I, x \in A_i\}.$$

Proposition 2.1.3. *Let $\{A_\alpha : \alpha \in I_A\}$ and $\{B_\beta : \beta \in I_B\}$ be families of subsets of a set X, then*

(1) Associativity:

$$\left(\bigcap_{\alpha \in I_A} A_\alpha\right) \cap \left(\bigcap_{\beta \in I_B} B_\beta\right) = \bigcap_{(\alpha,\beta) \in I_A \times I_B} (A_\alpha \cap B_\beta)$$

and

$$\left(\bigcup_{\alpha\in I_A} A_\alpha\right) \cup \left(\bigcup_{\beta\in I_B} B_\beta\right) = \bigcup_{(\alpha,\beta)\in I_A\times I_B} (A_\alpha \cup B_\beta)$$

(2) Distributivity:

$$A \cap \left(\bigcup_{\alpha\in I_A} A_\alpha\right) = \bigcup_{\alpha\in I_A} (A \cap A_\alpha)$$

and

$$A \cup \left(\bigcap_{\alpha\in I_A} A_\alpha\right) = \bigcap_{\alpha\in I_A} (A \cup A_\alpha)$$

(3) de Morgan's Laws:

$$\left(\bigcap_{\alpha\in I_A} A_\alpha\right)^c = \bigcup_{\alpha\in I_A} A_\alpha^c$$

and

$$\left(\bigcup_{\alpha\in I_A} A_\alpha\right)^c = \bigcap_{\alpha\in I_A} A_\alpha^c$$

Proof. We show how to prove (3). A common trick we use when proving the equality of 2 sets is to show that each set is the subset of the other set.

For $\subset$, $\forall a \in \left(\bigcap_{\alpha\in I_A} A_\alpha\right)^c$, since $a \in \bigcap_{\alpha\in I_A} A_\alpha \iff \forall \alpha \in I_A, a \in A_\alpha$, we have $a \in \left(\bigcap_{\alpha\in I_A} A_\alpha\right)^c \iff \exists \alpha_0 \in I_A, a \in A_{\alpha_0}^c$ by negation. Thus $a \in A_{\alpha_0}^c \subset \bigcup_{\alpha\in I_A} A_\alpha^c$.

For the other side, there exists an $\alpha_0 \in I_A$ such that $a \in A_{\alpha_0}^c$. Notice that $\bigcap_{\alpha\in I_A} A_\alpha \subset A_{\alpha_0}$, we have $A_{\alpha_0}^c \subset \left(\bigcap_{\alpha\in I_A} A_\alpha\right)^c$. $\square$

2.2 Relation and Operation

This subsection is to formally introduce relationship between elements in a set X. A binary relation R on a set X is a subset of the Cartesian product $X \times X$. For $(x, y) \in R$, we denote it as xRy.

There're several important types of relations.

Relation	Meaning
Reflexive	$\forall x \in X, xRx$
Symmetric	$xRy \implies yRx$
Transitive	$(xRy) \wedge (yRz) \implies (xRz)$
Complete	$\forall x, y \in X, (xRy) \vee (yRx)$

We now turn to some special relations.

2.2.1 Equivalence

We say a relation is a equivalence relation if it's reflexive, symmetric and transitive, a conventional notation for an equivalence relation is $\sim$. For each $x \in X$,

$$[x] := \{y \in X : x \sim y\}$$

is called the equivalence class of x and each $y \in [x]$ is called a representative of $[x]$. It follows to define

$$X/\sim := \{[x] : x \in X\}$$

called "X modulo $\sim$".

Before we proceed to give the definition of partition and quotient function, we calculate the example concerning function.

Example 2.2.1. Suppose $f : X \to f(X)$. We may define an equivalence relation as follows:

$$x \sim_f y \iff f(x) = f(y).$$

The equivalence class of x is given by the set $f^{-1}(f(x))$. We may add a restriction A on f such that $\tilde{f} := f|_A$ is a bijection, then

$$\bigcup_{x \in A} [\tilde{f}^{-1}(f(x))] = X$$

and $[\tilde{f}^{-1}(f(x))]$ is pairwise disjoint. Notice that $X/\sim_f = \{[x] : x \in A\}$. This means that $\operatorname{Im} f$ is partitioned by $[\tilde{f}^{-1}(f(x))]$ and we have a unique surjection $\pi : X \to X/\sim_f$ which maps $x \in X$ to $[x]$.

We will see the generality of how an equivalence relation partitions a set, before which we give the definition.

Definition 2.2.1 (Partition)**.** A partition of X is a subset $\mathcal{A}$ of 2^X such that $\mathcal{A}$ consists of pairwise dijoint subsets of X whose union is X.

Proposition 2.2.1. *Suppose $\sim$ is an equivalence relation, then $X/\sim$ is a partition of X.*

Proof. Notice that $\bigcup_{x\in X}[x] = X$. We only have to show that $\forall X_1, X_2 \in X/\sim$: $X_1 = X_2 \vee X_1 \cap X_2 = \emptyset$.

Suppose $[x] \neq [y]$ and $z \in [x] \cap [y]$, then by transitivity of $\sim$,

$$z \sim x \wedge y \sim z \implies x \sim y$$

contradicting with $[x] \neq [y]$. □

This leads to the former coonstruction of quotient functions.

Theorem 2.2.2. *For an equivalence relation $\sim$, there exists a unique surjection*

$$\pi : X \to X/\sim \qquad x \mapsto [x]$$

called the quotient function.

2.2.2 Order

A relation $\leq$ on a set X is a partial order if it's reflexive, transitive, and anti-symmetric*. We call the pair $(X, \leq)$ a partially ordered set. If, in addition to reflexiveness, transitivity, and anti-symmetric, completeness also holds for $\leq$, we say it's a total order and $(X, \leq)$ is a total ordered set.

We now turn to the elements in a partially ordered set $(X, \leq)$.

Definition 2.2.2 (Monotonicity)**.** Given a partially ordered set $(X, \leq)$, for $\{x_n\}_{n=1}^{\infty} \subset X$, we say it's **monotonically increasing** if $x_n \leq x_{n+1}, \forall n \in \mathbb{N}$; similarly, it's **monotonically decreasing** if $x_{n+1} \leq x_n, \forall n \in \mathbb{N}$.

* Anti-symmetry means $(x \leq y) \wedge (y \leq x) \implies x = y$

Definition 2.2.3 (Boundedness)**.** Given a partially ordered set $(X, \leq)$ and $A \subseteq X$, an element x is an **upper bound** of A if $a \leq x, \forall a \in A$. We say A is bounded above. And similarly, an element x is a **lower bound** of A if $x \leq a, \forall a \in A$ and it's bounded below.

Definition 2.2.4 (Supremum and Infimum)**.** Given a partially ordered set $(X, \leq)$ and $A \subseteq X$, if A is bounded above, then we define

$$\sup A = \min\{x \in X : x \text{ is an upper bound of } A\}$$

Similarly, if A is bounded below,

$$\inf A = \max\{x \in X : x \text{ is a lower bound of } A\}$$

Note 1. Note that, if $\sup A$ and $\max A$ exists, $\sup A \in A \iff \sup A = \max A$. Similarly, if $\inf A$ and $\min A$ exists, $\inf A \in A \iff \inf A = \min A$.

A very important example of partially order is the subset relation. We introduce the concept of lim inf and lim sup of a sequence of sets as an example, which is of great importance in measure theory.

Example 2.2.2. Let $\{A_i\}_{i=1}^{\infty}$ be a family of subsets of a set X. Let $\mathcal{A}^{(n)}$ and $\mathcal{A}_{(n)}$ denote

$$\mathcal{A}^{(n)} = \bigcap_{i=n}^{\infty} A_i, \quad \mathcal{A}_{(n)} = \bigcup_{i=n}^{\infty} A_i$$

and notice that

$$\mathcal{A}^{(n+1)} \subset \mathcal{A}^{(n)}, \mathcal{A}_{(n)} \subset \mathcal{A}_{(n+1)}, \quad \forall n \in \mathbb{N}$$

Since $\left(\left\{\mathcal{A}^{(n)}\right\}_{n=1}^{\infty}, \subset\right)$ and $\left(\left\{\mathcal{A}_{(n)}\right\}_{n=1}^{\infty}, \subset\right)$ are partially ordered sets, we have

$$\inf \left\{\mathcal{A}^{(n)}\right\}_{n=1}^{\infty} = \bigcup_{n=1}^{\infty} \mathcal{A}^{(n)} = \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i$$
$$\sup \left\{\mathcal{A}_{(n)}\right\}_{n=1}^{\infty} = \bigcap_{n=1}^{\infty} \mathcal{A}_{(n)} = \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i$$

denoted as $\liminf_{n\to\infty} A_n$ and $\limsup_{n\to\infty} A_n$ respectively.

The concept of partial order is important in economics as preference is actually a partial order. We can define certain kinds of preferences with ease.

Definition 2.2.5 (Rational Preference)**.** A preference is rational if and only if it's complete and transitive.

2.2.3 Mapping

We've encountered mappings before, where we say mapping is a rule that assigns each element of a set, the domain, to exactly one element of the image set. If treated carefully, we may find the word "rule" vague. We now try to give it a formal definition.

Consider the set $G = \{(x, f(x)) : x \in X\}$. To make $f : X \to Y$ a mapping, given any $x \in X$, there should exist only one $y \in Y$ such that $(x, y) \in G$. This leads to the following definition.

Definition 2.2.6 (Mapping)**.** A mapping $f : X \to Y$ is a binary relation

$$G = \{(x, f(x)) : x \in X \wedge (\exists! y \in Y \text{s.t.} f(x) = y)\}$$

and we denote $f(A) = \operatorname{Im} f|_A$, where $A \subset X$.

We now list some important concepts.

Definition 2.2.7. Let $f : X \to Y$ be a mapping.

Surjective f is surjective $\iff$ $\operatorname{Im} f = Y$.

Injective f is injective $\iff$ $(f(x) = f(y) \implies x = y)$.

Bijective f is bijective $\iff$ f is both injective and surjective.

Inverse Suppose f is bijective, then there exists an inverse function $f^{-1} : Y \to X$ such that $f^{-1} \circ f = \mathrm{Id}_X$.[†]

Proposition 2.2.3 (Set Valued Mapping)**.** *Suppose we have $f : X \to Y$ and families of subsets of X $\{A_i\}_{i \in I_A}$ and $\{B_j\}_{j \in I_B}$. Then we have*

[†] f^{-1} is well-defined as we'll show later that an operation has at most one identity element and then we can construct the inverse function with the bijective property. An important proposition is $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

(1) $f\left(\bigcup_{i\in I_A} A_i\right) = \bigcup_{i\in I_A} f(A_i)$
(2) $f\left(\bigcap_{i\in I_A} A_i\right) \subset \bigcap_{i\in I_A} f(A_i)$
(3) $f^{-1}\left(\bigcup_{j\in I_B} B_j\right) = \bigcup_{j\in I_B} f^{-1}(B_j)$
(4) $f^{-1}\left(\bigcap_{j\in I_B} B_j\right) = \bigcap_{j\in I_B} f^{-1}(B_j)$

2.2.4 Correspondence

For mappings, we assign only one element in the image set to each element in the domain. We now consider assigning multiple elements in the image set to each element in the domain, which is called correspondence.

Definition 2.2.8 (Correspondence)**.** A correspondence $\varphi : X \twoheadrightarrow Y$ is a binary relation

$$G = \{(x, \varphi(x)) : x \in X \wedge (\varphi(x) \subset Y)\}.$$

and for the subset A of X, we call $\varphi^u(A) := \{x \in X : \varphi(x) \subset A\}$ the upper inverse image of A; similarly, we call $\varphi^l(A) := \{x \in X : \varphi(x) \cap A \neq \emptyset\}$ the lower inverse image of A.

2.2.5 Operation

An operation is a mapping $\circledast : X \times X \to Y$ and we denote $\circledast(x, y)$ as $x \circledast y$ in convention. We say the subset A of X is closed under the operation $\circledast$ if $A \circledast A \subset A$.

Two important kinds of operation we often meet are associative and commutative operations.

Associative An operation $\circledast$ is associative if $(x \circledast y) \circledast z = x \circledast (y \circledast z)$ for all $x, y, z \in X$.

Commutative An operation $\circledast$ is commutative if $x \circledast y = y \circledast x$ for all $x, y \in X$.

Proposition 2.2.4 (Uniqueness of Identity Element)**.** *We call an element e an identity element if and only if $x \circledast e = e \circledast x = x$ for all $x \in X$. For a certain operation, there exsists at most one identity element.*

Proof. We prove by contradiction. Suppose there're two identity elements e and e'. Then $e = e \circledast e' = e'$, contradicting with $e \neq e'$. □

2.3 Countability

For finite sets, we can easily count the element and say it's countable. But what if the set have infinite many elements? Mathematicians believe that natural numbers can be counted and is the smallest infinite set. In this section we formally treat infinity, before which we introduce the axiom of choice.

2.3.1 The Axiom of Choice

Axiom 2.3.1 (the Axiom of Choice)**.** *For a family of non-empty sets $\{A_i\}_{i\in I}$, the Cartesian product $\prod_{i\in I} A_i$ is non-empty.*

That is equivalent to say there exists a choice function f such that $\forall i \in I, f(i) \in A_i$.

2.3.2 Cardinality

The cardinality of a finite set is defined as the number of elements, that is $|A| = \#A$. For infinite sets, the definition is complicated and we turn to compare the cardinality of 2 sets.

Definition 2.3.1 (Equality of Cardinality)**.** Two sets A and B are said to have the same cardinality if there exists a bijection from A to B, denoted as $|A| = |B|$. If there exists a subsection of B and a bijection from A to that subsection, we say $|A| \leq |B|$.

Proposition 2.3.1. *Let X, Y be sets, the following statements are equivalent:*

(1) $|X| \leq |Y|$
(2) there exists an injection from X to Y.
(3) there exists a surjection from Y to X.

Proof. (1) $\implies$ (2) is done by definition.

To show (2) $\implies$ (3), let g be a mapping from $f(X)$ to X such that $g \circ f|_{f(X)} = \mathrm{Id}_{f(X)}$,

which is a bijection. For a fixed $x \in X$, define

$$h(y) = \begin{cases} g(y), & y \in f(X) \\ x, & y \notin f(X) \end{cases}$$

which is a surjection from Y to X.

(3) $\implies$ (1) needs the axiom of choice. Let $A_x := \{y \in Y : f(y) = x\}$ and A_x is non-empty as f is surjective. By **axiom 2.3.1, the Axiom of Choice**, there exists a choice function h such that $h(x) \in A_x$. Notice that

$$x \neq x' \implies (A_x \cap A_{x'}) = \emptyset \implies h(x) \neq h(x')$$

Then h is an injection from X to Y. □

Actually, $(|\cdot|, \leq)$ is a total ordered set. See the following theorems.

Theorem 2.3.2 (Anti-symmetry and Completeness of Cardinality)**.** *Let X, Y be sets.*

Bernstein-Schroeder $(|Y| \leq |X| \wedge |X| \leq |Y|) \implies |X| = |Y|$

Completeness $|X| \leq |Y| \vee |Y| \leq |X|$

Note 2. Note that the completeness is equivalent to the axiom of choice.

2.3.3 Countable Set

Definition 2.3.2 (Countability)**.** A set X is countable if $|X| = |\mathbb{N}|$.

Theorem 2.3.3 (Existence of Countable Subsets)**.** *There exists an infinite countable subset for any infinite set.*

Proof. Let X be an infinite set. Denote $\mathcal{A}_n = \{A \subset X : |A| = 2^n\}$, $\forall n \in \mathbb{N}$. By **axiom 2.3.1, the Axiom of Choice**, there exists a family of non-empty sets $\{B_n\}_{n=1}^{\infty}$ such that $B_n \in \mathcal{A}_n$.

We now use **axiom 2.3.1, the Axiom of Choice**, again to construct a series $\{c_n\}_{n=1}^{\infty}$ such that $c_n \neq c_{n'}$. To do that, we construct a family of pairwise disjoint sets $\{C_n\}$

from B_n. An intuitive way to do so is to extract c_n from the new part of B_n, that is to construct $C_n = B_n \setminus \bigcup_{i=1}^{n-1} B_i$.

By construction, $f(n) : \mathbb{N} \mapsto c_n$ is an injection and by **proposition 2.3.1**, $|\mathbb{N}| \leq |\{c_n\}|$. Thus $\{c_n\}$ is a countable subset of X. □

Theorem 2.3.4. *A countable union of countable sets is countable.*

Proof. Let $\{A_n\}_{n=1}^{\infty}$ be a countable family of countable sets and denote $A_i = \{a_{ij}\}_{j=1}^{\infty}$. We try to find an injection from $\bigcup_{n=1}^{\infty} A_n$ to $\mathbb{N}$.

Suppose $\{A_n\}_{n=1}^{\infty}$ is pairwise disjoint, define $f : a_{mn} \mapsto m + \frac{(m+n-1)(m+n-2)}{2}$, the construction is illustrated below:

$$\begin{array}{ccccc}
a_{11} & a_{12} & \cdots & a_{1n} & \cdots \\
a_{21} & a_{22} & \cdots & a_{2n} & \cdots \\
\vdots & \vdots & \ddots & \vdots & \vdots \\
a_{m1} & a_{m2} & \cdots & a_{mn} & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots
\end{array}$$

Otherwise, denote $C_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$. Then C_n is a countable set and is pairwise disjoint. It's done with the result of the previous case. □

Proposition 2.3.5. *A finite product of countable sets is countable.*

Proof. Notice that $\mathcal{A}_1 \times \mathcal{A}_2 = \bigcup_{a \in \mathcal{A}_2} \{(a, a') : a' \in \mathcal{A}_1\}$, which is a countable union of countable sets. Therefore, $\prod_{i=1}^{n} \mathcal{A}_i$ is countable by the deductive relation $\prod_{i=1}^{n} \mathcal{A}_i = \prod_{i=1}^{n-1} \mathcal{A}_i \times \mathcal{A}_n$. □

Corollary 2.3.6. $\mathbb{N}^n$ *and* $\mathbb{Q}$ *is countable.*

Proof. Only have to show $\mathbb{Q}$ is countable. We can construct the surjection $f(m, n) = m/n$. Notice it maps $\mathbb{N}^2$ to $\mathbb{Q}$, thus $|\mathbb{Q}| \leq |\mathbb{N}^2| = |\mathbb{N}|$. Therefore, $\mathbb{Q}$ is countable by **theorem 2.3.2, Bernstein-Schroeder theorem**. $\square$

Chapter 3

Group, Ring and Field

With the concept of relation, we've added some structures to a set. For example, we've encountered $(\text{Funct}(X), \circ)$ and we know that it's associative, it has an identity element Id_X and for bijective ones, each element has an inverse.

In this chapter, we'll go deeper into understanding the abstract properties of a set and the added relation. We introduce some basic algebraic concepts and important propositions.

3.1 Group

3.1.1 the Definition

Definition 3.1.1 (Group)**.** If, for the 2-tuple $(G, \odot)$, the relation $\odot$ is associative, then we call a semi-group. Additionally, if there exists an identity element, we call it a monoid.

A group is a monoid such that for each element $g \in G$, there exists an inverse element $g^{-1} \in G$ and we call it an Abelian group if the relation $\odot$ is commutative.

Note 3. Note that the identity element is unique by **proposition 2.2.4, the uniqueness of identity element**. The following 2 propositions are also useful:

- $\forall g \in G, g \odot h = g \iff h \odot g = g$

- $(g \odot h)^{-1} = h^{-1} \odot g^{-1}$

Example 3.1.1. Drirect sum of $\mathcal{F}^n$

Definition 3.1.2 (Subgroup)**.** For a nonempty subset H of G, if it's closed under the operation $\odot$ * and $\forall h \in H, h^{-1} \in H$, we say $(H, \odot)$ is a subgroup of G†. We call $g \odot H$ and $H \odot g$ the left and right coset of $g \in G$ with respect to H and if $g \odot H = H \odot g$, we say H is a normal subgroup.

3.1.2 Homomorphisms

Now we turn to mappping one group to the other. A very special kind is the one that preserves the algebraic structure and we call it a homomorphism.

Definition 3.1.3 (Homomorphism)**.** A homomorphism φ is a mapping that maps $(G, \odot)$ to $(H, \circledast)$ such that $\forall g_1, g_2 \in G : \varphi(g_1 \odot g_2) = \varphi(g_1) \circledast \varphi(g_2)$. A homomorphism that maps G to itslef is called an endomorphism.

We list some important definitions and propositions.

Proposition 3.1.1. *A homomorphism maps the identity element to the identity element.*

Proof. Since $e_H \circledast \varphi(e_G) = \varphi(e_G \odot e_G) = \varphi(e_G) \circledast \varphi(e_G)$, we have $e_H = \varphi(e_G)$. □

Proposition 3.1.2. *A homomorphism maps the inverse element to the inverse of element.*

Proof. Notice that

$$\varphi(g) \circledast \varphi(g^{-1}) = \varphi(g \odot g^{-1}) = e_H \implies \varphi(g)^{-1} = \varphi(g^{-1}).$$

□

Definition 3.1.4 (Kernel)**.** The kernel of a homomorphism $\varphi : G \to H$ is defined by

$$\mathrm{Ker}(\varphi) := \varphi^{-1}(e_H) = \{g \in G : \varphi(g) = e_H\}$$

* Meaning

$$H \odot H \subset H :\iff \forall h_1, h_2 \in H : (h_1 \odot h_2) \in H.$$

† We can simply denote it as H if the operation is clear from the context.

and it's a normal subgroup of G[‡].

Proposition 3.1.3. *φ is injective $\iff$ $\mathrm{Ker}(\varphi)$ is trivial*[§].

Proof. ($\Rightarrow$) is trivial as injectivity makes $\varphi^{-1}(e_H)$ single-valued and by **proposition 3.1.1** we have $e_G \in \mathrm{Ker}(\varphi)$.

For ($\Leftarrow$), suppose there exist $g_1 \neq g_2$ such that $\varphi(g_1) = \varphi(g_2)$. Then $\varphi(e_G) = \varphi(g_2) \circledast \varphi(g_2^{-1}) = \varphi(g_1) \circledast \varphi(g_2^{-1}) = \varphi(g_1 \odot g_2^{-1}) = e_H$. This contradicts with $\mathrm{Ker}(\varphi) = \{e_G\}$ as $g_1 \odot g_2^{-1} \neq e_G$. □

3.1.3 Isomorphisms

Definition 3.1.5 (Isomorphism)**.** An isomorphism is a bijective homomorphism, denoted by $\cong$. If it maps a group to itself, we call it an automorphism.

Example 3.1.2 (The Set of All Automorphisms as a Group)**.** Let $\mathrm{Aut}(G)$ be the set of all automorphisms of G, then $(\mathrm{Aut}(G), \circ)$ is a group, which we call it an automorphism group.

Proposition 3.1.4 (Induced Operation)**.** *Let $(G, \odot)$ be a group and H is a nonempty set. Given a bijection $\varphi : G \to H$, we define the operation $\circledast$*

$$\mu \circledast \nu := \varphi(\varphi^{-1}(\mu) \odot \varphi^{-1}(\nu)), \mu, \nu \in H$$

then $(H, \circledast)$ is a group and φ is an isomorphism.

Proof. Showing that $(H, \circledast)$ is a group is trivial. To show that φ is a homomorphism, given $h_1, h_2 \in H$, there exist $g_1, g_2 \in G$ such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Then by definition, $h_1 \circledast h_2 = \varphi(g_1 \odot g_2)$. □

A very important result to know is that isomorphism hold exactly the same algebraic structure, which provides some convenience in calculation. For example, when it's hard to deal with $g_1 \odot g_2$, we may turn to $h_1 \circledast h_2$ and use φ^{-1} to get $g_1 \odot g_2$.

‡ To show this, first verify that $\mathrm{Ker}(\varphi)$ is indeed a subgroup. And it's normal as we notice that $\forall k \in g \odot \mathrm{Ker}(\varphi), \exists \mu \in \mathrm{Ker}(\varphi)$ such that $k = g \odot \mu$. Let $\nu = g \odot \mu \odot g^{-1}$, we have $\nu \in \mathrm{Ker}(\varphi)$ and since $\nu \odot g = g \odot \mu = k$, we have $k \in \mathrm{Ker}(\varphi) \odot g$. The other side can be proved similarly.

§ That is $\mathrm{Ker}(\varphi) = \{e_G\}$.

3.2 Ring and Field

In this section, we add another operations to a group. We require that the set forms an Abelian group with respect to one of the operations and we want a certain kind of distributivity. This leads us to ring. And we can further add more requirements to rings and make it a field.

3.2.1 The Definition

Definition 3.2.1 (Ring)**.** For a 3-tuple $(R, +, \cdot)$, if

(1) $(R, +)$ is an Abelian group[¶],
(2) $(R, \cdot)$ is a semi-group,
(3) Distributivity:

$$a \cdot (b + c) = a \cdot b + a \cdot c, \quad (a + b) \cdot c = a \cdot c + b \cdot c, \qquad \forall a, b, c \in R$$

With more specific requirements, we have different kinds of rings:

Commutative Ring If $\cdot$ is commutative, then $(R, +, \cdot)$ is a commutative ring.

Ring with Unity If there exists an identity element $1_R \in R$ for multiplication, then $(R, +, \cdot)$ is a ring with unity.

The definition of subring is similar to that of subgroup.

Definition 3.2.2 (Subring)**.** Let S be a nonempty subset of a ring $(R, +, \cdot)$. If S is a subgroup of $(R, +)$ and is closed under multiplication, we say it's a subring.

Remark 1. Given the existence of additive and multiplicative identity elements in a ring, we know that they're unique. Also, there may exists $a \neq 0_R$ such that $a \cdot b = 0_R$ or $b \cdot a = 0_R$ for some $b \neq 0_R$. Such elements are called **zero divisors**.

3.2.2 Homomorphism and Isomorphism

The definition here is similar to that of group homomorphism and isomorphism.

¶ A conventional notation for the identity element is 0_R.

Definition 3.2.3 (Ring Homomorphism and Isomorphism)**.** Let R, S be rings. A mapping $\varphi : R \to S$ is a ring homomorphism if it preserve the multiplication and addition operations. If φ is bijective, then it is a ring isomorphism.

Note that if φ is a ring homomorphism, then it's a group homomorphism for $(R, +)$ and $(S, +)$ and the kernel is defined by $\mathrm{Ker}(\varphi) = \{r \in R : \varphi(r) = 0_S\}$.

3.2.3 Ordered Field

In this subsubsection, we add an order relation to a field and make it an ordered field.

Definition 3.2.4 (Ordered Field)**.** An ordered field is a field $(F, +, \cdot)$ equipped with a total order relation $\leq$, if the following statements hold for all $a, b, c \in F$:

(1) If $a \leq b$, then $a + c \leq b + c$.
(2) If $0 \leq a$ and $0 \leq b$, then $0 \leq a \cdot b$.

3.3 Polynomials

Part II

Vector Space and Linear Algebra

Chapter 4

Vector Space

From this chapter we start to formally introduce linear algebra. We'll see that linear algebra is similar to a ring but with some fine properties so that we can achieve more powerful results. To study linear algebra, we start with the vector space and gradually add more structures to it.

4.1 the Definition of Vector Space and Subspaces

Definition 4.1.1 (Vector Space)**.** A vector space over a field $\mathcal{F}$ is a triple $(V, +, \cdot)$ consisting of a nonempty set V whose elements are called **vectors**, and two operations called **vector addition** and **scalar multiplication** respectively, defined as

$$+ : (u, v) \in V^2 \mapsto u + v, \quad \cdot : (a, v) \in \mathcal{F} \times V \mapsto a \cdot v$$

which satisfy the following axioms:

(1) $(V, +)$ is an Abelian group.
(2) Identity element: $\exists 1 \in \mathcal{F}$ such that $\forall v \in V, 1 \cdot v = v$.
(3) Associativity: $\forall a, b \in \mathcal{F}, \forall v \in V, (ab) \cdot v = a \cdot (b \cdot v)$.
(4) Distributivity: $\forall a \in \mathcal{F}, \forall u, v \in V, a \cdot (u + v) = a \cdot u + a \cdot v$ and $\forall a, b \in \mathcal{F}, \forall v \in V, (a + b) \cdot v = a \cdot v + b \cdot v$.

We can define the concept of subspace similar to that of subgroup and subring.

Definition 4.1.2 (Subspace of a Vector Space)**.** Let U be a nonempty subset of a $\mathcal{F}$-vector space $(V, +, \cdot)$. If U is closed under vector addition and scalar multiplication, then we say U is a subspace of V.

4.2 Operation of Subspaces

4.2.1 Sum

Definition 4.2.1 (Sum)**.** Let $V_1, \cdots, V_m$ be vector spaces over the same field $\mathcal{F}$. We define their sum by

$$V_1 + V_2 + \cdots + V_m := \{v_1 + v_2 + \cdots + v_m : v_i \in V_i, i = 1, 2, \ldots, m\}.$$

Remark 2. Note that $V_1 + V_2 + \cdots + V_m$ is also a subspace of V and is the smallest subspace containing all $V_1, \cdots, V_m$.

4.2.2 Direct Sum

For an element v in the sum of subspaces, suppose that

$$v = v_1 + v_2 + \cdots + v_m = u_1 + u_2 + \cdots + u_m.$$

Then

$$(v_1 - u_1) + \cdots + (v_m - u_m) = 0.$$

The most special case is when $v_i = u_i$ $(\forall i \in \{1, 2, \ldots, m\})$. This special case is defined as a direct sum.

Definition 4.2.2 (Direct Sum of Subspaces)**.** If every element in the sum of subspaces can be represented uniquely, then the sum is called the **direct sum** of the subspaces, denoted by $V_1 \oplus V_2 \oplus \cdots \oplus V_m$.

Theorem 4.2.1 (Equivalent definition of direct sum)**.** *$V_1 + \cdots + V_m$ is a direct sum $\iff$ the equation $v_1 + v_2 + \cdots + v_m = 0$ holds only when $v_i = 0$ for all i.*

Proof. It follows directly from the above discussion. □

For the case of 2 subspaces, we have a more specific result.

Theorem 4.2.2 (Direct Sum of 2 Subspaces)**.** *Let U and W be subspaces of V. Then $U + W$ is a direct sum if and only if $U \cap W = \{0\}$.*

Proof. ($\Rightarrow$) For any $u \in U \cap W$, there exists $(-u) \in W$ such that $0 = u + (-u)$. Since the representation is unique, we must have $u = 0$.

($\Leftarrow$) Suppose $u \in U$, $w \in W$ satisfy $u + w = 0$, i.e., $u = -w \in W$. Then $u \in U \cap W$, which implies $u = w = 0$. By the equivalent definition, $U + W$ is a direct sum. □

4.3 Subspace Spanned by Vectors

4.3.1 Span

We now turn to the elements of the vector space. For any $v \in V$, we can find a list of $a_1, \cdots, a_n \in \mathcal{F}$ and $v_1, \cdots, v_n \in V$ such that $v = a_1v_1 + \cdots + a_nv_n$. In this way, given a list of vectors, we can construct a vector space by generating vectors with all $(a_1, \cdots, a_n) \in \mathcal{F}^n$. We call such a space a **span**, defined by

$$\operatorname{span}(v_1, \cdots, v_n) := \{a_1v_1 + \cdots + a_nv_n : a_i \in \mathcal{F}, v_i \in V\}.$$

If $\operatorname{span}(v_1, \cdots, v_n) = V$, we say that $v_1, \cdots, v_n$ spans V.

Notice that $x^0, \cdots, x^n$ can span $\mathcal{P}_n(\mathcal{F})$ and $x^0, \cdots, x^n, 2x^n$ can do it as well. This inspires us to find out which is the smallest one that can span a vector space.

4.3.2 Linear Independence

In this section, we introduce the concept of linear independence and we'll find out that the smallest spanning list is linearly independent. Consider the elements in the spanned subspace. Let

$$v = a_1v_1 + \cdots + a_mv_m = b_1v_1 + \cdots + b_mv_m,$$

then

$$(a_1 - b_1)v_1 + \cdots + (a_m - b_m)v_m = 0.$$

The most special case is when $a_i = b_i$ (for all $i \in \{1, \ldots, m\}$). To characterize this situation, we introduce linear independence.

Definition 4.3.1 (Linear Independence)**.** If $a_1v_1 + \cdots + a_mv_m = 0$ holds only when $a_i = 0, \forall i \in \{1, \cdots, m\}$, then the vectors $v_1, \ldots, v_m$ are said to be **linearly independent**.

This following propositions directly follow from the above discussion.

Proposition 4.3.1. *Removing vectors from a linearly independent list does not change the linear independence of the list.*

Proposition 4.3.2. *The vectors $v_1, \ldots, v_m$ are linearly independent if and only if every element in* $\operatorname{span}(v_1, \ldots, v_m)$ *can be represented uniquely.*

We know proceed to study the length of a spanning list and a linearly independent list and we first do some preperations. Intuitively, a spanning list should be longer than a linearly independent list. The following lemma formalizes this idea.

Lemma 4.3.3. *Suppose $v_1, \ldots, v_m$ are linearly dependent. Then there exists $k \in \{1, 2, \ldots, m\}$ such that $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$. Furthermore, removing v_k does not change the spanned subspace, i.e.,*

$$\operatorname{span}(v_1, \ldots, v_{k-1}, v_{k+1}, \ldots, v_m) = \operatorname{span}(v_1, \ldots, v_m).$$

Proof. By definition, we have $a_1v_1 + \cdots + a_mv_m = 0$ where $a_1, \cdots, a_m$ are not all zeros. Suppose that $a_{k+1} = \cdots = a_m = 0$ and $a_1a_2 \cdots a_k \neq 0$. We have $v_k = -\frac{a_1}{a_k}v_1 - \cdots - \frac{a_{k-1}}{a_k}v_k$, implying that $v_k \in \operatorname{span}(v_1, \cdots, v_{k-1})$.

For any vector in $\operatorname{span}(v_1, \ldots, v_m)$, it can be expressed as the linear combination of $v_1, \ldots, v_m$. By substituting the expression of v_k into this linear combination, we can express it solely in terms of $v_1, \ldots, v_{k-1}, v_{k+1}, \ldots, v_m$. Thus, the spanned subspace remains unchanged. □

Theorem 4.3.4. *Let $v_1, \ldots, v_m$ be a spanning list of a vector space V. Then any linearly independent list of vectors in V has length at most m.*

Proof. Let $u_1, \cdots, u_m$ be a linearly independent list. Consider adding u_1 to $v_1, \cdots, v_n$ and by **lemma 4.3.3** we know $u_1, v_1, \cdots, v_n$ is linearly dependent and we can remove

one of the vectors without changing the spanned space. Notice that we can choose one vector from $v_1, \cdots, v_n$ since u_1 is none-zero.

We now proceed to add u_2 to the previous list. Since $u_2 \notin \text{span}(u_1)$ by linear independence, we can choose a vector from the v's and remove it. We continue doing so until no vectors are left in the linearly independent list. We now get a list with n vectors which contains $u_1, \cdots, u_m$, implying $m \leq n$. □

4.4 Finite Dimensional Vector Space

In the previous section, we've discovered that the smallest spanning list is linearly independent. This inspire us to present a vector space with the help of a linearly independent spanning list. In this section, we move on to discover the properties of the subspace spanned by a linearly independent list with finite length.

4.4.1 Basis

Definition 4.4.1 (Basis)**.** A **basis** of a vector space V is a linearly independent list of vectors that spans V.

The most important result is the following, which shows how to construct a basis.

Lemma 4.4.1 (Steinitz)**.** *Suppose $v_1, \ldots, v_m$ is a spanning list of a vector space V, then there exists a subset of $\{v_1, \ldots, v_m\}$ that forms a basis of V. Similarly, if $u_1, \ldots, u_n$ is a linearly independent list in V, then there exists a list $u_{n+1}, \ldots, u_m$ in V such that $u_1, \ldots, u_m$ forms a basis of V.*

Proof. For the first part, if $v_1, \ldots, v_m$ is linearly independent, then it is a basis. Otherwise, by **lemma 4.3.3**, we can remove one vector without changing the spanned space. Continuing this process, we will eventually obtain a linearly independent list that spans V, which is a basis.

For the second part, if $u_1, \ldots, u_n$ spans V, then it is a basis. Otherwise, there exists $v \in V$ such that $v \notin \text{span}(u_1, \ldots, u_n)$. We can add v to the list to form $u_1, \ldots, u_n, v$. Repeating this process, we will eventually obtain a spanning list that contains $u_1, \ldots, u_n$. By the first part, we can extract a basis from this spanning list. □

4.4.2 Dimension

It's natural to define the length of the basis as the dimension of the vector space. Before we present the definition, we need to show that the length of the basis is unique.

Theorem 4.4.2. *Any two basis of a vector space V have the same length.*

Proof. Suppose B_1 and B_2 are two basis of V. We can view B_1 as a spanning list and B_2 as a linearly independent list. By **theorem 4.3.4**, we know that the length of B_2 is at most that of B_1. By symmetry, the length of B_1 is at most that of B_2. Thus, they have the same length. □

Definition 4.4.2 (Dimension)**.** The **dimension** of a vector space V, denoted by $\dim V$, is defined as the length of any basis of V.

Theorem 4.4.3 (Criterion for Basis of a Finite-dimensional Vector Space)**.** *A spanning or linearly independent list of a vector space V is a basis if and only if its length equals* $\dim V$.

Proof. It's done with **lemma 4.4.1, the Steinitz exchange lemma**. □

We close this subsection with the following results concerning operations of vector spaces and their dimension.

Theorem 4.4.4 (the Dimension Inequality)**.** *For vector space V and its subspace U, suppose $U \subset V$. Then $0 \leq \dim U \leq \dim V$ and it becomes an equation only when U is trivial or $U = V$.*

Theorem 4.4.5. *For vector space V and U,*

$$\dim(U + V) = \dim U + \dim V - \dim(U \cap V).$$

Theorem 4.4.6. *Let $V_1, \cdots, V_m$ be vector spaces, then $V_1 + \cdots + V_m$ is a direct sum* $\iff \dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + V_m$.

Chapter 5

Linear Mapping and Matrix

By definition, for a vector sapce V, we know $(V, +)$ is an Abelian group. It's natural to consider the group homomorphisms and isomorphisms on vector spaces. We'll see linear mapping as a group homomorphism and discover some more fine properties in this chapter.

5.1 Linear Mapping

5.1.1 the Definitions

Definition 5.1.1 (Linear Mapping)**.** For a function φ from the vector space V to the vector space W, if the following statements hold, we call it a linear mapping*.

- $\varphi(v_1 + v_2) = \varphi(v_1) + \varphi(v_2), \forall v_1, v_2 \in V$.
- $\varphi(\lambda v) = \lambda\varphi(v), \forall \lambda \in \mathcal{F}$.

We denote the set of all linear mappings from V to W as $\mathcal{L}(V, W)$.

It's clear that a linear mapping is indeed a homomorphism, so we can easily translate some of the definitions and propositions from group homomorphisms to linear mappings.

* One may ask if such a mapping exists. The answer is yes as there exists a unique linear mapping that takes the basis of V to the basis of W, see (Axler, 2024, p. 54) for proof of this proposition.

Definition 5.1.2 (Kernel)**.** The kernel of a linear mapping $\varphi \in \mathcal{L}(V, W)$ is defined as

$$\operatorname{Ker} \varphi = \{v \in V : \varphi(v) = 0\}$$

Proposition 5.1.1. *Suppose $\varphi \in \mathcal{L}(V, W)$, then the following holds:*

- $\varphi(0) = 0$
- *φ is injective $\iff$ $\operatorname{Ker}(\varphi)$ is trivial.*

We can also define the operations on $\mathcal{L}(V, W)$.

Definition 5.1.3 (Operations on $\mathcal{L}(V, W)$)**.** Let $\mathcal{L}(V, W)$ be the set of all linear mappings from V to W, we have 3 operations on it:

Addition $\forall \varphi, \varpi \in \mathcal{L}(V, W), (\varphi + \varpi)(v) = \varphi(v) + \varpi(v)$

Scalar Multiplication $\forall \lambda \in \mathcal{F}, (\lambda\varphi)(v) = \lambda(\varphi(v))$

Product (Combination) $\forall \varphi, \varpi \in \mathcal{L}(V, W), (\varphi\varpi)(v) = \varphi \circ \varpi(v)$

Then $\mathcal{L}(V, W)$ is a vector space.

5.1.2 Isomorphic Linear Mapping

Before we start to deal with isomorphisms, we first dig deeper into the linear mappings of finite-dimensional vector spaces.

Theorem 5.1.2 (Fundamental Theorem of Linear Mappings)**.** *Suppose $\varphi \in \mathcal{L}(V, W)$, then*

$$\dim V = \dim \operatorname{Im} \varphi + \dim \operatorname{Ker} \varphi$$

Proof. Let $u_1, \cdots, u_n$ be a basis of $\operatorname{Ker} \varphi$. By **lemma 4.4.1, the Steinitz exchange lemma**, we can extend them as $u_1, \cdots, u_n, v_1, \cdots, v_m$ so that it becomes a basis of V. We close the proof by showing $\varphi(v_1), \cdots, \varphi(v_n)$ is a basis of $\operatorname{Im} \varphi$.

We first show that $\varphi(v_1), \cdots, \varphi(v_n)$ spans $\operatorname{Im} \varphi$. Notice that $\forall v \in V$, we can rewrite it as $a_1u_1 + \cdots + a_mu_m + b_1v_1 + \cdots + b_nv_n = v$. Thus

$$\varphi(v) = b_1\varphi(v_1) + \cdots + b_n\varphi(v_n)$$

implying $\varphi(v) \in \operatorname{span}(\varphi(v_1), \cdots, \varphi(v_n))$.

Now we show the linear independence[†]. Suppose $c_1v_1 + \cdots + c_nv_n = 0$, then $\varphi(c_1v_1 + \cdots + c_nv_n) = 0$, meaning $c_1v_1 + \cdots + c_nv_n \in \operatorname{Ker}\varphi$. We can rewrite it with the linear combination of $u_1, \cdots, u_m$, say $c_1v_1 + \cdots + c_nv_n = d_1u_1 + \cdots + d_mu_m$. Since $u_1, \cdots, u_n, v_1, \cdots, v_m$ is linearly independent, all c's and d's are zero. □

Corollary 5.1.3. *Suppose V and W are vector spaces. If $\dim V < \dim W$, $\varphi \in \mathcal{L}(V, W)$ is NOT surjective. If $\dim V > \dim W$, $\varphi \in \mathcal{L}(V, W)$ is NOT injective*

With these results, we can depict an isomorphism with ease.

Definition 5.1.4 (Isomorphic Linear Mapping)**.** If a linear mapping is bijective, we say it's isomorphic.

We know by **corollary 5.1.3** that a necessary condition for a linear mapping to be isomorphic is that the finite-dimensional vector spaces have the same dimension.

Corollary 5.1.4. *Suppose $\varphi \in \mathcal{L}(V, W)$ with $\dim V = \dim W$, Then*

$$\varphi \text{ is isomorphic} \iff \operatorname{Ker}\varphi \text{ is trivial} \iff \operatorname{Im}\varphi = W \iff \varphi \text{ is invertible}$$

Proof. Use **theorem 5.1.2, the fundamental theorem of linear mappings**, we have

$$\varphi \text{ is surjective} \iff \operatorname{Im}\varphi = W \iff \dim \operatorname{Ker}\varphi = 0 \iff \varphi \text{ is injective}$$

□

If there exists an isomorphism from V to W, we say V and W are isomorphic.

Theorem 5.1.5. *V and W are isomorphic $\iff \dim V = \dim W$.*

5.2 Matrix

We have already explored the abstract concept of linear mappings and we know that isomorphic linear mappings are of great importance as they maintain the structure

[†] From this part we can also show that an isomorphic linear mapping maps a linearly independent list to a linearly independent list.

of a certain vector space. We now turn to a more concrete representation of linear mappings, that is matrix, before which we consider a specific vector space, $\mathcal{F}^n$.

Example 5.2.1 ($\mathcal{F}^n$ as a Vector Space)**.** For a field $\mathcal{F}$, you may verify that the direct product $\mathcal{F}^n$‡ is a $\mathcal{F}$-vector space with the group addition and scalar multiplication defined by $\cdot : (\lambda, (f_1, \cdots, f_n)) \in \mathbb{F} \times \mathcal{F}^n \mapsto (\lambda f_1, \cdots, \lambda f_n)$. We have $\dim \mathcal{F}^n = n$ and its basis is given by

$$\underbrace{(1, 0, \cdots, 0), (0, 1, \cdots, 0), \cdots, (0, 0, \cdots, 1)}_{n \text{ vectors}}.$$

Proof. We show the basis part. It's clear that these n vectors span $\mathcal{F}^n$. Now we show the linear independence. Suppose $a_1(1, 0, \cdots, 0) + a_2(0, 1, \cdots, 0) + \cdots + a_n(0, 0, \cdots, 1) = 0$, then we have $(a_1, a_2, \cdots, a_n) = 0$, implying all a_i's are zero. □

Note that we can also write $(f_1, \cdots, f_n)$ in a column, that is

$$\begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix}$$

or simply denote it by $(f_1, \cdots, f_n)^t$. Wether to write a vector in $\mathcal{F}^n$ in a row or in a column is basically for simplicity of the notation. We'll see the benefits when presenting the m-direct product of $\mathcal{F}^n$. Suppose $\mathbf{f}_1, \cdots \mathbf{f}_m \in \mathcal{F}^n$, elements of the direct product $\mathcal{F}^{n,m}$ can be written as $(\mathbf{f}_1, \cdots \mathbf{f}_m)$ and we drop the brackets of $\mathbf{f}$'s, that is

$$\begin{pmatrix} f_{11} & \cdots & f_{1m} \\ \vdots & \ddots & \vdots \\ f_{n1} & \cdots & f_{nm} \end{pmatrix}.$$

A similar result of the previous example is that $\mathcal{F}^{n,m}$ is a vector space with dimension nm.

‡ What we mean here is to see the field $\mathcal{F}$ as a group $(\mathcal{F}, +)$.

5.2.1 From Vector to Coordinate

From **theorem 5.1.5**, we know that V, whose dimension is n, and $\mathcal{F}^n$ is isomorphic. That is to say there exists a bijection $\mathcal{M} : V \to \mathcal{F}^n$. For a basis of V, denoted by $v_1, \cdots, v_n$, we construct the isomorphism as follows:

$$v_i \mapsto (\underbrace{0, \cdots, 1, \cdots, 0}_{\text{the } i\text{th slot equals to } 1}).$$

Then, for $v \in V$, we can write it as $v = a_1 v_1 + \cdots + a_n v_n$ and

$$\mathcal{M}(v) = a_1 \mathcal{M}(v_1) + \cdots + a_n \mathcal{M}(v_n)$$

this leads us to the definition of coordinate.

Definition 5.2.1 (Coordinate)**.** Suppose for $v \in V$, we have $v = a_1 v_1 + \cdots + a_n v_n$, then its coordinate is defined as

$$\mathcal{M}(v) = (a_1, a_2, \cdots, a_n).$$

5.2.2 From Linear Mapping to Matrix

Notice that $\dim \mathcal{L}(V, W) = nm$ if $\dim V = m$ and $\dim W = n$. So it's natural to construct an isomorphism to give a specific representation of linear mappings. Suppose $\varphi \in \mathcal{L}(V, W)$ with $\dim V = n$ and $\dim W = m$. Let $v_1, \cdots, v_m$ be a basis of V and $w_1, \cdots, w_n$ be a basis of W. We try to calculate $\mathcal{M}(\varphi(v_i))$ so that we can know the coordinate of any vectors in $\varphi(V)$ by linear combination. Suppose $\varphi(v_i) = a_{1i} w_1 + a_{2i} w_2 + \cdots + a_{ni} w_n$, then

$$\mathcal{M}(\varphi(v_i)) = (a_{1i}, a_{2i}, \cdots, a_{ni})^t.$$

Thus, we can present φ by

$$\begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{pmatrix}.$$

Definition 5.2.2 (Matrix)**.** Suppose $\varphi \in \mathcal{L}(V, W)$ and the basis of V and W are

$(v_1, \cdots, v_m)$ and $(w_1, \cdots, w_n)$ respectively, then the matrix of φ with respect to the basis $(v_1, \cdots, v_m)$ and $(w_1, \cdots, w_n)$ is defined as

$$\mathcal{M}(\varphi, (v_1, \cdots, v_m), (w_1, \cdots, w_n))^{\S} = \begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & a_{1k} & & \\ & & \vdots & & \\ & & a_{nk} & & \end{pmatrix} \end{array}$$

where $\{a_{ij}\}_{(i,j)=(1,1)}^{(n,m)}$ satisfy $\varphi(v_k) = a_{1k}w_1 + a_{2k}w_2 + \cdots + a_{nk}w_n$ for $k = 1, \ldots, m$. We denote the set of matrix with n rows and m columns as $\mathcal{F}^{n,m}$.

5.2.3 Operations of Matrix

As mentioned before, a matrix is basically a linear mapping. In this way we define the operations of matrixes analogous to the operations of linear mappings.

Addition Suppose $\varphi, \varpi \in \mathcal{L}(V, W)$ and we have $\varphi(v_k) = a_{1k}w_1 + \cdots + a_{nk}w_n$ and $\varpi(v_k) = b_{1k}w_1 + \cdots + b_{nk}w_n$.

$$(\varphi + \varpi)(v_k) = \varphi(v_k) + \varpi(v_k) = (a_{1k} + b_{1k})w_1 + \cdots + (a_{nk} + b_{nk})w_n$$

Scalar Multiplication Suppose $\varphi \in \mathcal{L}(V, W)$ and $\varphi(v_k) = a_{1k}w_1 + \cdots + a_{nk}w_n$. For $\lambda \in \mathbb{F}$,

$$(\lambda\varphi)(v_k) = \lambda\left(\varphi(v_k)\right) = \lambda a_{1k}w_1 + \cdots + \lambda a_{nk}w_n$$

Product (Composition) Suppose $\varphi \in \mathcal{L}(V, W)$ and $\psi \in \mathcal{L}(W, U)$, with $\varphi(v_k) = a_{1k}w_1 + \cdots + a_{nk}w_n$ and $\psi(w_k) = b_{1k}u_1 + \cdots + b_{pk}u_p$. Then the

§ Simply use the notation $\mathcal{M}(\varphi)$ when the basis is clear from the context.

composition $\psi \circ \varphi$ satisfies:

$$\begin{aligned}(\psi \circ \varphi)(v_k) = \psi\left(\varphi(v_k)\right) &= a_{1k}\psi(w_1) + \cdots + a_{nk}\psi(w_n) \\ &= a_{1k}\sum_{i=1}^{p} b_{i1}u_i + \cdots + a_{nk}\sum_{i=1}^{p} b_{in}u_i \\ &= \left(\sum_{j=1}^{n} b_{1j}a_{jk}\right)u_1 + \cdots + \left(\sum_{j=1}^{n} b_{pj}a_{jk}\right)u_p\end{aligned}$$

This leads us to the following definitions.

Definition 5.2.3 (Operations of Matrix)**.** In the space of matrices, we can define the following operations:

- **Addition:** For matrices $A = \mathcal{M}(\varphi)$ and $B = \mathcal{M}(\varpi)$ in $\mathcal{F}^{n,m}$ (with respect to the same bases),

$$\mathcal{M}(\varphi + \varpi) = \mathcal{M}(\varphi) + \mathcal{M}(\varpi),$$

that is,

$$\begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array}\!\!\begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & & & \\ & & a_{1k} & & \\ & & \vdots & & \\ & & a_{nk} & & \end{pmatrix} \end{array} + \begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array}\!\!\begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & & & \\ & & b_{1k} & & \\ & & \vdots & & \\ & & b_{nk} & & \end{pmatrix} \end{array}$$

$$= \begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array}\!\!\begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & & & \\ & & a_{1k} + b_{1k} & & \\ & & \vdots & & \\ & & a_{nk} + b_{nk} & & \end{pmatrix} \end{array}$$

- **Scalar Multiplication:** For a matrix $A = \mathcal{M}(\varphi)$ in $\mathcal{F}^{n,m}$ and a scalar $\lambda \in \mathbb{F}$,

$$\mathcal{M}(\lambda\varphi) = \lambda\mathcal{M}(\varphi),$$

that is,

$$\lambda \begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & a_{1k} & & \\ & & \vdots & & \\ & & a_{nk} & & \end{pmatrix} \end{array} = \begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & \lambda a_{1k} & & \\ & & \vdots & & \\ & & \lambda a_{nk} & & \end{pmatrix} \end{array}$$

- **Multiplication (Composition):** For matrices $A = \mathcal{M}(\varphi) \in \mathcal{F}^{n,m}$ (with respect to bases of V and W) and $B = \mathcal{M}(\psi) \in \mathcal{F}^{p,n}$ (with respect to bases of W and U),

$$\mathcal{M}(\psi \circ \varphi) = \mathcal{M}(\psi)\mathcal{M}(\varphi) = BA,$$

that is,

$$\begin{array}{c} \\ u_1 \\ \vdots \\ u_p \end{array} \begin{array}{c} \begin{array}{ccccc} w_1 & \cdots & w_k & \cdots & w_n \end{array} \\ \begin{pmatrix} & & b_{1k} & & \\ & & \vdots & & \\ & & b_{pk} & & \end{pmatrix} \end{array} \cdot \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & a_{1k} & & \\ & & \vdots & & \\ & & a_{nk} & & \end{pmatrix} \end{array} \begin{array}{c} \\ w_1 \\ \vdots \\ w_n \end{array}$$

$$= \begin{array}{c} \\ u_1 \\ \vdots \\ u_p \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_m \end{array} \\ \begin{pmatrix} & & \sum_{j=1}^n b_{1j} a_{jk} & & \\ & & \vdots & & \\ & & \sum_{j=1}^n b_{pj} a_{jk} & & \end{pmatrix} \end{array}$$

Note that the product matrix BA is in $\mathcal{F}^{p,m}$, corresponding to the linear mapping $\psi \circ \varphi : V \to U$.

Remark 3. There're several ways to understand product oa matrix. Let Col_k and Row_k denote the kth column and row respectively. Consider the matrix multiplication $C = AB$:

- entry of matrix product equals row times column: $c_{ij} = \mathrm{Row}_i(A) \cdot \mathrm{Col}_j(B)$.
- column(row) of matrix product equals matrix times column(row): $\mathrm{Col}_k(C) = A\,\mathrm{Col}_k(B)$.
- matrix multiplication as linear combinations of columns(rows): Notice that, suppose b is a vector presented by $(b_1, \cdots, b_n)$, $Ab = \sum_{i=1}^n b_i\,\mathrm{Col}_i(A)$. Thus, if we

see columns of B as vectors,

$$\mathrm{Col}_k(C) = \sum_{i=1}^{n} b_{ik}\,\mathrm{Col}_i(A).$$

The third one leads to the CR factorization, which plays a crucial role in the definition of rank.

5.2.4 Change of Basis

The definition implies that matrix may change if we choose a diiferent basis. In this section, we're going to find a formula for changing the basis, which can help us find the simplist matrix to present a certain linear mapping.

Before the matrix version, we consider changing the basis of a vector. Let $e_1, \ldots, e_n$ be a basis of V. For any $\alpha \in V$, we write $\alpha = \lambda_1 e_1 + \cdots + \lambda_n e_n$, which can be represented as an $n \times 1$ matrix (column vector):$\alpha = (\lambda_1, \cdots, \lambda_n)'$

Let $f_1, \ldots, f_n$ be another basis of V. We investigate how to express α with respect to the new basis f, starting from its representation in the basis e. Write each f_i in terms of e:

$$f_i = \sum_{k=1}^{n} a_{ik} e_k, \quad i = 1, 2, \ldots, n.$$

Now suppose α is expressed in the f-basis as $\alpha = \mu_1 f_1 + \cdots + \mu_n f_n$. Then

$$\begin{aligned}
\alpha &= \mu_1 \left(\sum_{k=1}^{n} a_{1k} e_k \right) + \cdots + \mu_n \left(\sum_{k=1}^{n} a_{nk} e_k \right) \\
&= \left(\sum_{k=1}^{n} \mu_k a_{k1} \right) e_1 + \cdots + \left(\sum_{k=1}^{n} \mu_k a_{kn} \right) e_n \\
&= \left(a_{11}, \ldots, a_{n1} \right) \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix} e_1 + \cdots + \left(a_{1n}, \ldots, a_{nn} \right) \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix} e_n \\
&= \begin{pmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}.
\end{aligned}$$

Thus, the coordinates of α in the e-basis are obtained by multiplying the transition matrix by its coordinate vector in the f-basis.

Chapter 6

Normed Vector Space

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Chapter 7

Multilinear Form and Determinant

One good way to introduce determinant is through solving equation systems and define it by deduction. Consider solving

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

then we have

$$x_1 = \frac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{12}a_{21}}, x_2 = \frac{b_2a_{11} - b_1a_{21}}{a_{11}a_{22} - a_{12}a_{21}}$$

and denote

$$\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

We can rewrite the solution as

$$x_1 = \frac{\det\begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}}, x_2 = \frac{\det\begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}}$$

Now we turn to the 3-dimensional case:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

Similarly, by elimination method, we have

$$x_1 = \frac{b_1 \det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12}\det\begin{pmatrix} b_2 & a_{23} \\ b_3 & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} b_2 & a_{22} \\ b_3 & a_{32} \end{pmatrix}}{a_{11}\det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12}\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}}$$

$$x_2 = \frac{a_{11}\det\begin{pmatrix} b_2 & a_{23} \\ b_3 & a_{33} \end{pmatrix} - b_1\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} a_{21} & b_2 \\ a_{31} & b_3 \end{pmatrix}}{a_{11}\det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12}\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}}$$

$$x_3 = \frac{a_{11}\det\begin{pmatrix} a_{22} & b_2 \\ a_{32} & b_3 \end{pmatrix} - a_{12}\det\begin{pmatrix} a_{21} & b_2 \\ a_{31} & b_3 \end{pmatrix} + b_1\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}}{a_{11}\det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12}\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}}$$

This leads us to denote

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}\det\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12}\det\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13}\det\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}$$

and rewrite the solution as

$$x_1 = \frac{\det\begin{pmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}, x_2 = \frac{\det\begin{pmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}, x_3 = \frac{\det\begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}}$$

In this way can we define determinant in a deductive way and achieve the important Cramer's Law easily.

Definition 7.0.1. Suppose $A = (a_{ij})_{n\times n}$, the determinant of A is defined as

$$\det A = \sum_{i=1}^{n} (-1)^{i+1} a_{1i} \det\begin{pmatrix} a_{22} & \cdots & a_{2,i-1} & a_{2,i+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & & \vdots \\ a_{n2} & \cdots & a_{n,i-1} & a_{n,i+1} & \cdots & a_{nn} \end{pmatrix}$$

However, this definition arises only as a notation, lacking the geometric understanding of determinants. To do this, we start from the concept of **multilinear form**.

7.1 Bilinear Form

A bilinear form on V is a function on $V \times V$ and is linear in each slot while holding the other one fixed.

Definition 7.1.1 (Bilinear Form)**.** Let V be a vector space over field $\mathbb{F}$. A function $\beta : V \times V \to \mathbb{F}$ is called a bilinear form on V if

$$v \mapsto \beta(u, v), \quad u \mapsto \beta(u, v)$$

are both linear mapping on V. We denote $V^{(2)}$ as the set of all bilinear forms in V.

An example of bilinear form is the inner product.

Example 7.1.1 (Inner Product)**.** Let β be the inner product $\langle \cdot, \cdot \rangle$. Recall the linearity of inner product in each slot and we can verify by definition that inner product is indeed a bilinear form.

7.1.1 Symmetric and Alternating Bilinear Form

However, though the inner product is symmetric, that is $\beta(u, v) = \beta(v, u)$, symmetry does not always hold for bilinear forms. There is a special kind of bilinear form called **symmetric bilinear form**.

Definition 7.1.2 (Symmetric Bilinear Form)**.** A bilinear form β from $V^{(2)}$ is called symmetric if

$$\beta(u, v) = \beta(v, u), \quad \forall u, v \in V.$$

We denote $V^{(2)}_{sym}$ as the set of all symmetric bilinear forms in $V^{(2)}$. It's trivial to show that $V^{(2)}_{sym}$ is a subspace of $V^{(2)}$.

Another special and important kind of bilinear form is **alternating bilinear form**. We'll see its importance when we reach the final theorem that $V^{(2)}$ can be separated as the direct sum of $V^{(2)}_{sym}$ and $V^{(2)}_{alt}$.

Definition 7.1.3 (Alternating Bilinear Form)**.** A bilinear form β from $V^{(2)}$ is called alternating if

$$\beta(v, v) = 0, \quad \forall v \in V.$$

We denote $V^{(2)}_{alt}$ as the set of all alternating bilinear forms in $V^{(2)}$. It's trivial to show that $V^{(2)}_{alt}$ is a subspace of $V^{(2)}$.

Proposition 7.1.1 (Characterization of Alternating Bilinear Form)**.** *A bilinear form α is alternating if and only if*

$$\alpha(u, v) = -\alpha(v, u), \quad \forall u, v \in V.$$

Proof. Notice that, by definition of bilinear form, we have

$$\alpha(u + v, u + v) = \alpha(u, u) + \alpha(u, v) + \alpha(v, u) + \alpha(v, v)$$

Suppose α is alternating, then $\alpha(u + v, u + v) = \alpha(u, v) + \alpha(v, u) = 0$, which implies

$\alpha(u,v) = -\alpha(v,u)$.

Conversely, suppose $\alpha(u,v) = -\alpha(v,u)$, then $\alpha(v,v) = -\alpha(v,v)$, which implies $\alpha(v,v) = 0$ for all v. Therefore, α is alternating. □

Now we turn to the final theorem of alternating and symmetric bilinear forms.

Theorem 7.1.2 (Decomposition of Bilinear Form)**.** *Given a vector space V, we have*

$$V^{(2)} = V^{(2)}_{sym} \oplus V^{(2)}_{alt}.$$

Proof. We first show that $V^{(2)} = V^{(2)}_{sym} + V^{(2)}_{alt}$. Suppose $\beta \in V^{(2)}$, Let

$$\beta_{sym} = \frac{\beta(u,v) + \beta(v,u)}{2}, \quad \beta_{alt} = \frac{\beta(u,v) - \beta(v,u)}{2}$$

It's trivial that $\beta_{sym} \in V^{(2)}_{sym}$ and $\beta_{alt} \in V^{(2)}_{alt}$, and $\beta = \beta_{sym} + \beta_{alt}$.

Then we only have to show that $V^{(2)}_{sym} \cap V^{(2)}_{alt} = \{0\}$. Suppose $\beta \in V^{(2)}_{sym} \cap V^{(2)}_{alt}$. Then β is both symmetric and alternating, so $\beta(u,v) = \beta(v,u)$ and $\beta(u,v) = -\beta(v,u)$, which implies $\beta(u,v) = 0$ for all u, v. Therefore, $V^{(2)}_{sym} \cap V^{(2)}_{alt} = \{0\}$. □

7.1.2 Matrix of a Bilinear Form

The definition is similar to the definition of matrix of a linear mapping.

Definition 7.1.4 (Matrix of a Bilinear Form)**.** Suppose $\beta \in V^{(2)}$ and $e_1, \cdots, e_n$ is a basis. Then the matrix of β is defined as $\{\beta(e_i, e_j)\}^n_{i,j=1}$.

The operation of bilinear forms can be represented by the operation of their matrix. The addition and scalar multiplication is similar to the case of linear mapping. The following theorem shows the case of composition.

Theorem 7.1.3. *Let*

$$\beta_\varphi(v,w) = \beta(\varphi(v), w) \qquad \beta^\varphi(v,w) = \beta(v, \varphi(w))$$

Then

$$\mathcal{M}(\beta_\varphi) = \mathcal{M}(\varphi)'\mathcal{M}(\beta) \qquad \mathcal{M}(\beta^\varphi) = \mathcal{M}(\beta)\mathcal{M}(\varphi)$$

The change of basis formular also holds if we use the transpose instead of the inverse.

Theorem 7.1.4 (Change-of-basis Formular)**.** *Suppose $e_1, \cdots, e_n$ and $f_1, \cdots, f_n$ are two bases of V and let $C = \mathcal{M}(I, (e_1, \cdots, e_n), (f_1, \cdots, f_n))$. Then for any $\beta \in V^{(2)}$, we have*

$$\mathcal{M}(\beta, (e_1, \cdots, e_n)) = C^T \mathcal{M}(\beta, (f_1, \cdots, f_n))C$$

Suppose $\beta \in V^{(2)}_{sym}$. We know by definition that

$$\beta(e_i, e_j) = \beta(e_j, e_i) \tag{7.1}$$

for all i, j. For a matrix satifying this condition, we call it a symmetric one.

Definition 7.1.5 (Symmetric Matrix)**.** A matrix A is called symmetric if $A' = A$.

We know that the matrix of a symmtric bilinear form is a symmetric matrix by (7.1) and it has nothing to do with the choice of basis. The converse statement also holds.

Theorem 7.1.5. *Suppose $\beta \in V^{(2)}$. The followings are equivalent:*

(1) β is symmetric.
(2) The matrix of β is symmetric with respect to any basis.
(3) The matrix of β is symmetric with respect to some basis.

Proof. We've already shown that (1) $\implies$ (2). It's trivial that (2) $\implies$ (3). We now show that (3) $\implies$ (1). Suppose $\mathcal{M}(\beta, (e_1, \cdots, e_n)) = \{a_{ij}\}$ with $a_{ij} = a_{ji}$ for all i, j. For any $u = \sum_{i=1}^n u_i e_i$ and $v = \sum_{i=1}^n v_i e_i$, we have

$$\begin{aligned}
\beta(u, v) &= \sum_{i=1}^n \sum_{j=1}^n u_i v_j \beta(e_i, e_j) \\
&= \sum_{i=1}^n \sum_{j=1}^n u_i v_j a_{ij} \\
&= \sum_{i=1}^n \sum_{j=1}^n u_i v_j a_{ji} = \beta(v, u)
\end{aligned}$$

This shows that β is symmetric. □

Notice that any diagonal matrix is symmetric, we're inspired to consider the relationship between symmetric bilinear forms and diagonal matrices. The following theorem shows that we can always find a basis such that the matrix of a symmetric bilinear form is diagonal.

Theorem 7.1.6. *Suppose $\beta \in V^{(2)}$, then $\beta \in V^{(2)}_{sym} \iff \exists(e_1, \cdots, e_n)$ such that*

$$\mathcal{M}(\beta, (e_1, \cdots, e_n))$$

is diagonal. And moreover, if V is a real inner product space, then we can find an orthonormal basis such that the matrix of β is diagonal.

So what are the diagonal elements? We can show that the diagonal elements are exactly the eigenvalues of β.

7.1.3 Quadratic Form

7.2 Multilinear Form

We now turn to define determinants and this subsection is for preparation. The definition of multilinear form is a natural extension of bilinear form.

Definition 7.2.1 (Multilinear Form)**.** Let V be a vector space over field $\mathbb{F}$. A k-multilinear form

$$\beta : \underbrace{V \times V \times \cdots \times V}_{k \text{ times}} \to \mathbb{F}$$

satisfies that

$$v_i \mapsto \beta(v_1, \ldots, v_i, \ldots, v_k)$$

are all linear mapping on V for each $i = 1, 2, \ldots, k$. We denote $V^{(k)}$ as the set of all k-multilinear forms in V.

7.2.1 Alternating Multilinear Form

For multilinear forms, we only concern about alternating ones.

Definition 7.2.2 (Alternating Multilinear Form)**.** A multilinear form β from $V^{(k)}$ is

called alternating if

$$\beta(v_1, \ldots, v_k) = 0$$

whenever $v_i = v_j$ for some $i \neq j$. We denote $V_{alt}^{(k)}$ as the set of all alternating multilinear forms in $V^{(k)}$. It's trivial to show that $V_{alt}^{(k)}$ is a subspace of $V^{(k)}$.

By linearity at each slot, it's trivial to show that for a linearly dependent set $\{v_1, \ldots, v_k\}$, we have $\alpha(v_1, \ldots, v_k) = 0$ for any alternating multilinear form $\alpha \in V_{alt}^{(k)}$. The counterpositive satement implies that if $\alpha(v_1, \ldots, v_k) \neq 0$ for some alternating multilinear form $\alpha \in V_{alt}^{(k)}$, then $\{v_1, \ldots, v_k\}$ is linearly independent. Actually, with the theorem $\dim V_{alt}^{(\dim V)} = 1$, which will be shown later, the converse statement also holds.

Theorem 7.2.1. *For a nonzero $\alpha \in V_{alt}^{(k)}$, $\alpha(v_1, \ldots, v_k) \neq 0 \iff \{v_1, \ldots, v_k\}$ is linearly independent.*

7.2.2 Swapping the Entries

Theorem 7.2.2 (Swapping 2 Slots of an Alternating Multilinear Form)**.** *For alternating multilinear form α and distinct $i, j \in \{1, 2, \ldots, k\}$, we have*

$$\alpha(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -\alpha(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k)$$

Proof. We show the case of swapping v_1 and v_2. Notice that

$$\begin{aligned}
0 &= \alpha(v_1 + v_2, v_1 + v_2, \ldots, v_k) \\
&= \alpha(v_1, v_1, \ldots, v_k) + \alpha(v_1, v_2, \ldots, v_k) + \alpha(v_2, v_1, \ldots, v_k) + \alpha(v_2, v_2, \ldots, v_k) \\
&= \alpha(v_1, v_2, \ldots, v_k) + \alpha(v_2, v_1, \ldots, v_k)
\end{aligned}$$

We have $\alpha(v_1, v_2, \ldots, v_k) = -\alpha(v_2, v_1, \ldots, v_k)$. □

To generalize the swapping theorem, we consider swapping multiple slots and multiple times. The intuitive idea asks us to count how many times we just swapped and we can use the above theorem. That is to consider, given the original list $(v_1, \ldots, v_k)$ and the swapped list $(v_{j_1}, \ldots, v_{j_k})$ with exactly the same elements but in different orders, how many swaps are made. We don't know how many swaps are made since there're multiple ways to do it and actually, how the swaps were

made wouldn't change the result. So we can reconstruct the process of swapping by doing adjacent swaps, that is only swapping v_i and v_{i+1}. In this manner, we can count the swap by comparing the original order and the final order. We introduce the concept of permutation to understand the full picture following our intuition.

Definition 7.2.3 (Permutation and its Sign)**.** A permutation of the list $(1, \ldots, k)$ is an element of $\operatorname{perm} k := \{(j_1, \ldots, j_k) \in \mathbb{N}^k : j_u \neq j_v \quad \text{and} \quad 1 \leq j_i \leq k\}$.

The sign of a permutation maps a certain element of $\operatorname{perm} k$ to either $+1$ or -1. For a permutation $(j_1, \ldots, j_k) \in \operatorname{perm} k$, its sign $\operatorname{sign}(j_1, \ldots, j_k)$ is defined as

$$\operatorname{sign}(j_1, \ldots, j_k) = (-1)^N$$

where N is the cardinality of $\{(m, n) : 1 \leq m < n \leq k \text{ and } j_m > j_n\}$.

The next result is straightforward.

Proposition 7.2.3 (Generalization of the Swapping)**.** *Given a permutation* $(j_1, \ldots, j_k) \in \operatorname{perm} k$ *and an alternating multilinear form* $\alpha \in V_{alt}^{(k)}$*, we have*

$$\alpha(v_{j_1}, \ldots, v_{j_k}) = \operatorname{sign}(j_1, \ldots, j_k)\alpha(v_1, \ldots, v_k)$$

7.3 Determinant

7.3.1 From Alternating Multilinear Form to Determinant

We now turn to multilinear forms in $V_{alt}^{(\dim V)}$. The following 2 theorems are of the most significance in defining determinants and calculating them.

Theorem 7.3.1 (Calculation of Alternating $\dim V$-linear Forms)**.** *Let* $\dim V = n$ *and* $e_1, \cdots, e_n$ *is a basis of* V*. For any vectors* $v_1, \cdots, v_n$*, suppose* $v_i = b_{1i}e_1 + \cdots + b_{ni}e_n$*, then*

$$\alpha(v_1, \ldots, v_n) = \alpha(e_1, \ldots, e_n) \sum_{(j_1, \cdots, j_n) \in \operatorname{perm} n} \operatorname{sign}(j_1, \ldots, j_n) b_{j_1 1} \cdots b_{j_n n}.$$

Proof. Notice that

$$\begin{aligned}\alpha(v_1,\ldots,v_n) &= \sum_{j_1=1}^{n} b_{j_1 1}\alpha(e_{j_1},\ldots,v_n)\\ &= \cdots = \sum_{j_1=1}^{n}\sum_{j_2=1}^{n}\cdots\sum_{j_n=1}^{n} b_{j_1 1}b_{j_2 2}\cdots b_{j_n n}\alpha(e_{j_1},\ldots,e_{j_n})\\ &= \alpha(e_1,\ldots,e_n)\sum_{(j_1,\cdots,j_n)\in \operatorname{perm} n}\operatorname{sign}(j_1,\ldots,j_n)b_{j_1 1}\cdots b_{j_n n}.\end{aligned}$$

where the last equation holds by **proposition 7.2.3**. □

Theorem 7.3.2. *Let* $\dim V = n$*, then* $\dim V_{alt}^{(n)} = 1$.

Proof. Suppose $\alpha \neq 0$ and for a basis $e_1,\cdots,e_n$, we have $\alpha(e_1,\cdots,e_n) \neq 0$. For any $\alpha' \in V_{alt}^{(n)}$, there exists $c \in \mathcal{F}$ such that $\alpha'(e_1,\cdots,e_n) = c\alpha(e_1,\cdots,e_n)$. For any list $v_1,\cdots,v_n \in V$, use **theorem 7.3.1, fomular of alternating** $\dim V$**-linear forms**, we have

$$\begin{aligned}\alpha'(v_1,\ldots,v_n) &= \alpha'(e_1,\ldots,e_n)\sum_{(j_1,\cdots,j_n)\in \operatorname{perm} n}\operatorname{sign}(j_1,\ldots,j_n)b_{j_1 1}\cdots b_{j_n n}\\ &= c\alpha(v_1,\ldots,v_n)\end{aligned}$$

meaning α and α' are linearly dependent, thus $\dim V_{alt}^{(n)} \leq 1$.

We now show that there exists a nonzero $\alpha \in V_{alt}^{(n)}$. □

We can construct a new alternating $\dim V$-linear form with a linear mapping $\varphi \in \mathcal{L}(V)$. Consider

$$\alpha_\varphi(v_1,\cdots,v_n) := \alpha(\varphi(v_1),\cdots,\varphi(v_n)).$$

Since $\dim V_{alt}^{(\dim V)} = 1$, there exists a unique number $D \in \mathcal{F}$ such that $\alpha_\varphi = D\alpha$. We call the unique number determinant of an operator.

Definition 7.3.1 (Determinant)**.** Suppose $\varphi \in \mathcal{L}(V)$, the unique number such that $\alpha_\varphi = (\det\varphi)\alpha$ is called the determinant of φ. This notation can also be applied to matrix, let A denote the matrix of φ, define $\det A := \det\varphi$.

Note 4. Note that this definition hasn't done yet. Recall that the definition of a matrix has something to do with the choice of basis. We may first choose the standard basis of $\mathcal{F}^n$ and after showing the multiplicative property of determinants, we will know this definition is well-defined by the change-of-basis formular.

7.3.2 Calculation of Determinants

By the next theorem we can view determinants as alternating multilinear forms and then apply **proposition 7.2.3** to calculate them.

Theorem 7.3.3 (Determinant as Alternating Multilinear Form)**.** *Suppose $v_1, \cdots, v_n \in \mathcal{F}^n$ and $(v_1, \cdots, v_n)$ is a $n \times n$ matrix. The mapping $(v_1, \cdots, v_n) \mapsto \det(v_1, \cdots, v_n)$ is an alternating multilinear form.*

Proof. Let $e_1, \cdots, e_n$ be the standard basis of $\mathcal{F}^n$ and α be an alternating n-linear form in $\mathcal{F}^n$ such that $\alpha(e_1, \cdots, e_n) = 1$. Suppose $(v_1, \cdots, v_n)$ is the matrix of φ with respect to the standard basis. Then for any list $v_1, \cdots, v_n$ in $\mathcal{F}^n$,

$$\begin{aligned}\det(v_1, \cdots, v_n) &= \det \varphi \\ &= (\det \varphi)\alpha(e_1, \cdots, e_n) \\ &= \alpha(\varphi(e_1), \cdots, \varphi(e_n)) = \alpha(v_1, \cdots, v_n)\end{aligned}$$

then $\det(v_1, \cdots, v_n)$ is alternating since α is alternating. □

By **proposition 7.2.3**, we can calculate a determinant as follows.

Theorem 7.3.4 (Formular of Determinants)**.** *Suppose $A = (a_{ij})_{n\times n}$, then*

$$\det A = \sum_{(j_1, \cdots, j_n) \in \operatorname{perm} n} \operatorname{sign}(j_1, \ldots, j_n) a_{j_1 1} \cdots a_{j_n n}.$$

One may verify the equivalence between the deductive way and the above way.

7.3.3 Property of Determinants

Chapter 8

Eigenvalue and Eigenvector

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Part III

Metric Space and Topology Space

Chapter 9

Metric Space and Convergence

9.1 Metric Space

The introduction of metric space is to formally state the concept of distance. Consider the distance between 2 points, we naturally require it to be non-negative and it equals to 0 if and only if the two points are exactly the same. It should also be symmetric, meaning the distance of A to B is the same as the one of B to A. Moreover, the triangle inequality should also hold, which is a formal illustration of "a straight line is the shortest distance between 2 points". This leads to the definition of metric.

9.1.1 the Definitions

Definition 9.1.1 (Metric Space)**.** Let X be a set and we define the metric, a mapping from $X \times X \to \mathbb{R}+$, such that

(1) $d(x, y) = 0 \iff x = y$.
(2) Symmetry: $\forall x, y \in X : d(x, y) = d(y, x)$.
(3) Triangle inequality: $\forall x, y, z \in X : d(x, y) \leq d(x, z) + d(z, y)$.

Remark 4. The third axiom can be extended as

$$\forall x, y, z \in X : d(x, y) \geq |d(x, z) - d(z, y)|.$$

This holds by noticing that $d(x,y) \geq d(x,z) - d(y,z)$ and interchanging x and y, then

$$d(x,y) = d(y,x) \geq d(y,z) - d(x,z) = -[d(x,z) - d(y,z)]$$

This leads to the result.

We show some special metrics.

Example 9.1.1 (the Discrete and Natural Metric)**.** We call the following metrics discrete metric and natural metric respectively,

$$d_D : (x,y) \in X \times X \mapsto \delta_{x,y} \qquad d_N : (x,y) \in X \times X \mapsto |x-y|$$

where $\delta_{x,y}$ means

$$\delta_{x,y} = \begin{cases} 0, & x = y \\ 1, & x \neq y \end{cases}$$

9.1.1.1 Diameter

The previous defition is concerning 2 points, we now extend it to the sets.

Definition 9.1.2 (Diameter)**.** For a subset Y of X,

$$\operatorname{diam}(Y) := \sup_{x,y \in Y} d(x,y).$$

If it's finite, we say Y is d-bounded or simply bounded when the metric is clear form the context.

9.1.1.2 Neighborhood

We introduce the concept of neighborhood for a metric space, before which we introduce ball. We'll see the general definition of neighborhood when encontering topology.

Definition 9.1.3 (Ball)**.** An open ball with center at a and radius r is defined as

$$\mathbb{B}(a,r) := \{x \in X : d(a,x) < r\}$$

and a closed ball with center at a and radius r is defined as

$$\bar{\mathbb{B}}(a, r) := \{x \in X : d(a, x) \leq r\}$$

Definition 9.1.4 (Neighborhood)**.** We say $U \subset X$ is a neighborhood of $a \in X$ if $\exists r \in \mathbb{R} : \mathbb{B}(a, r) \subset U$. And we denote $\mathcal{U}(a)$ as the set of all neighborhoods of a.

9.1.1.3 Cluster Point

We can define cluster point for a sequence.

Definition 9.1.5 (Cluster Point of a Sequence)**.** Suppose $a \in X$, we call it a cluster point of $\{x_n\}_{n=1}^{\infty}$ if $\forall U \in \mathcal{U}(a) : U$ contains infinitely many terms of $\{x_n\}_{n=1}^{\infty}$.

The following proposition is straightforward by definition.

Proposition 9.1.1. *The followings are equivalent:*

(1) a is a cluster point of $\{x_n\}_{n=1}^{\infty}$.
(2) $\forall U \in \mathcal{U}(a)$ and $m \in \mathbb{N} : \exists n > m$ such that $x_n \in U$.
(3) $\forall \varepsilon \in \mathbb{R}$ and $m \in \mathbb{N} : \exists n > m$ such that $x_n \in \mathbb{B}(a, \varepsilon)$.

This can be extended to a set as well.

Definition 9.1.6 (Cluster Point of a Set)**.** Let $E \subset X$ and we say x_0 is a cluster point of E if $\forall \delta \in \mathbb{R}^+ : \exists x \in E \setminus \{x_0\}$ such that $d(x_0, x) < \delta$. Or equivalently, $\forall \delta \in \mathbb{R}^+ : \exists r \in \mathbb{R}$ such that $(E \setminus \{x_0\}) \cap \mathbb{B}(x_0, r) \neq \emptyset$.

9.1.2 Induced Metric

9.2 Convergence

9.2.1 Convergence of a Sequence

What is convergence? An intuitive explaination is that when n goes large enough, the sequence gets close enough to a certain number a. We can use the distance to describe "close enough".

Definition 9.2.1 (Convergence of a Sequence)**.** For a sequence $\{x_n\}_{n=1}^{\infty} \subset X$, suppose there exists $x \in X$ such that

$$\forall \varepsilon > 0 : \exists N = N(\varepsilon) \in \mathbb{N} \text{ such that } d(x_n, x) < \varepsilon, \forall n > N.^*$$

In such case we say x is the limit of $\{x_n\}$, denoted by $x_n \to x(n \to \infty)$ or $\lim_{n\to\infty} x_n = x$.

The following proposition is trivial by definition.

Proposition 9.2.1. *Suppose $\{x_n\} \subset X$, the followings are equivalent:*

(1) $x_n \to x(n \to \infty)$
(2) Each neighborhood of x contains almost all terms of $\{x_n\}$, that is the complement of each neighborhood contains at most finite terms of $\{x_n\}$.
(3) $\forall U \in \mathcal{U}(x) : \exists N = N(U)$ *such that* $x_n \in U, \forall n > N$.

Remark 5. It's clear that a limit is a cluster point, but a cluster point is not necessarily a limit. It's sometimes easier to show a sequence doesn't converge to a by showing a is not a cluster point.

The following 2 propositions are proved using the triangle inequality.

Proposition 9.2.2 (Every Convergent Sequence is Bounded)**.** *Suppose $x_n \to x(n \to \infty)$, then $\{x_n\}$ is bounded.*

Proof. By definition, there exists $N \in \mathbb{N}$ such that $\forall n > N : d(x_n, x) < 1$. Thus,

$$d(x_n, x_m) < d(x_n, x) + d(x_m, x) < 2, \quad \forall n, m \geq N$$

Let $M := \max_{i,j\leq N}\{2, d(x_i, x_j)\}$, then $2M$ is a bound by noticing that $d(x_n, x_m) < d(x_n, x_N) + d(x_m, x_N)$. □

Proposition 9.2.3 (Uniqueness of Limit)**.** *Suppose $x_n \to x(n \to \infty)$, then x is the unique cluster point of $\{x_n\}$.*

* Or equivalently, $x_n \in \mathbb{B}(x, \varepsilon), \forall n > N$.

Proof. Suppose there exists x' such that $x_n \to x(n \to \infty)$. We try to construct a ball which contains finite terms of $\{x_n\}$.

Let $\epsilon := d(x, x')/2$. Since $x_n \to x(n \to \infty)$, we have $\exists N = N(\epsilon) : d(x, x_n) < \epsilon, \forall n > N$ dy definition. Then

$$\forall n > N : d(x_n, x') \geq |d(x, x_n) - d(x, x')| > \epsilon$$

meaning there exists at most finitely many terms of $\{x_n\}$ in $\mathbb{B}(x', \epsilon)$. □

Corollary 9.2.4. *Every convergent sequence has a unique limit.*

With this corollary, we show our definition is well-defined.

9.3 Improper Convergence

From now on we narrow it down and consider sequences defined on real lines.

We calculate an example to show how to categorize sequence in terms of convergence and divergence.

Example 9.3.1. Consider the following 3 sequences.

$$a_n = \frac{1}{n}, \quad b_n = n, \quad c_n = \begin{cases} a_n, & n = 2m \\ a_n + 1, & n \neq 2m \end{cases}$$

It's trivial that a_n converge to 0 and $a_n + 1$ converge to 1. This leads to c_n being divergent since a convergent sequence should have a unique limit. However, we can't find a suitable limit for b_n.

9.3.1 Infinite Limit

In this subsection, we're to extend the metric on $\mathbb{R}$ so that sequences like b_n can be given a limit.

Definition 9.3.1 (Extended Concepts Concerning $\pm\infty$)**.** Let $\bar{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$, the extended concepts are defined as:

Neighborhood a subset U is called a neighborhood of $+\infty$ if there exists $K \in \mathbb{R}^+$ such that $(K, +\infty) \subset U$ (or $(-\infty, -K)$ for $-\infty$).

Cluster Point $\pm\infty$ is called the cluster point of $\{x_n\}$ if each neighborhood of $\pm\infty$ contains infinitely many terms of $\{x_n\}$.

Convergence to $\pm\infty$(Improper Convergence) $x_n \to \pm\infty (n \to \infty)$ is defined as:

$$\forall \varepsilon > 0 : \exists K \in \mathbb{R}^+ \text{ such that } |x_n| \geq \varepsilon, \forall n > K.$$

9.3.2 Some Special Sequences

9.3.2.1 Monotone Sequence

A sequence is monotonic if $\forall n \in \mathbb{N} : x_n \leq (\geq) x_{n+1}$.

Theorem 9.3.1 (Monotone and Bounded)**.**

Theorem 9.3.2 (Stolze)**.**

9.3.2.2 Subsequence

Theorem 9.3.3 (Bolzano-Weierstrass)**.**

9.3.3 Limit Superior and Limit Inferior

This subsection is to describe sequences like b_n, with which we can give it a limit.

Definition 9.3.2 (Limit Superior and Limit Inferior)**.** We construct the following 2 sequences:

$$\bar{x}_n := \sup_{k \geq n} x_n \qquad \underline{x}_n := \inf_{k \geq n} x_n$$

We know by definition that they are monotonic, thus converging in $\bar{\mathbb{R}}$, whose limit is defined by

$$\limsup_{n \to \infty} x_n := \lim_{n \to \infty} \bar{x}_n \qquad \liminf_{n \to \infty} x_n := \lim_{n \to \infty} \underline{x}_n.$$

The next theorem shows how limit inferior and superior dipict cluster points.

Theorem 9.3.4 (Limit Inferior(Superior) is the Smallest(Greatest) Cluster Point)**.** *Let*

x_* and x^* be the smallest and greatest cluster point. Then

$$\limsup_{n\to\infty} x_n = x^* \qquad \liminf_{n\to\infty} x_n = x_*.$$

Proof. □

Suppose $\{x_n\}$ converges in $\mathbb{R}$, it must have a unique cluster point, meaning $\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n$. We'll show in the next theorem that the converse statement also holds.

Theorem 9.3.5. *Suppose $\{x_n\}$ is a sequence in $\mathbb{R}$, then*

$$x_n \to x(n \to \infty) \iff \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = x.$$

Proof. □

9.4 Completeness

9.4.1 Cauchy Sequence

9.4.2 Banach Space

Chapter 10

Topology Space

Topology space concerns the structure of open sets. The first understanding of a open set is from the open ball we defined in the metric space chapter, where we also define the concept of neighborhood. We'll see that most of the definitions arise from previous concepts, this is how we generalize things. However, the topology not necessarily relies on metric. We'll see the generality of open sets and return to the topology induced by a metric, where we'll see some interestong results.

10.1 the Definition

Definition 10.1.1 (Topology)**.** Given a set X and a family of subsets τ. We say τ is a topology on X if the followings hold:

(1) $\emptyset, X \in \tau$.
(2) τ is closed under any union.
(3) τ is closed under finite intersection.

Then we say τ is a topology and (X, τ) is a topology space.

Example 10.1.1. We define the following 2 topologies:

$$\tau_{trivial} := \{\emptyset, X\} \qquad \tau_{discrete} := 2^X$$

which is called the trivial topology and the discrete topology.

Notice that $\tau_{trivial} \subset \tau_{discrete}$, we say $\tau_{discrete}$ is (strictly) finer. It's not always the case that we can compare two topology.

10.1.1 Open Set and Closed Set

As mentioned before, elements of τ are called open sets and we define that a closed set has an open complement.

We can now generalize the definition of neighborhood.

Definition 10.1.2 (Neighborhood)**.** Suppose $x \in A \subset X$, if there exists an open set U such that $x \in U \subset A$, then we say A is a neighborhood of x. If A is an open set itself, we say A is an open neighborhood of x.

Using the language of neighborhood, we can characterize open sets in a diferent way.

Theorem 10.1.1. *Suppose A is a subset of X, then A is open $\iff \forall x \in A$: there exists a neighborhood U_x of x such that $U_x \subset A$.*

Proof. ($\Rightarrow$) This side is trivial since A is a neighborhood of each x of X.

($\Leftarrow$) For this side, notice that by definition, there exists an open set $O_x \subset U_x$ for each $x \in X$ such that $x \in O_x$, then

$$A = \bigcup_{x \in A} \{x\} \subset \bigcup_{x \in A} O_x \subset \bigcup_{x \in A} U_x \subset A$$

implying $\bigcup_{x \in A} O_x = A$ and it's open. □

We now turn to closed set, by De-Morgan's law, we know that the set $\mathcal{F}$ of all closed sets satisfies:

(1) $\emptyset, X \in \mathcal{F}$
(2) $\mathcal{F}$ is closed under any intersection.
(3) $\mathcal{F}$ is closed under finite union.

Similarly, we can generalize the concept of cluster point.

Definition 10.1.3 (Limit Point(Accumulation Point))**.** Suppose X is a topology space and A is a subset of X. We say $x \in X$ is a limit point of A if for any neighborhood U_x of x, we have $U_x \cap (A \setminus \{x\}) \neq \emptyset$. And we denote the set of all limit points of A as A', called the derived set of A.

Theorem 10.1.2. *Suppose X is a topology space and A is a subset of X. We have A is closed $\iff A' \subset A$.*

Proof. We'll see this theorem is a simple corollary of **theorem 10.1.5**. □

10.1.2 Interior, Closure and Boundary

Given a set in X, it may be open, closed or neither open nor closed. If open/ closed, its structure is clear. If neither, we hope we can decompose it into the union of an open set and a closed set so that we can have a full picture of the structure of the set. To do that, we first define the interior and closure of a set.

Definition 10.1.4 (Interior and Closure)**.** Suppose (X, τ) is a topology space and $A \subset X$, the interior of A, A°, is the biggest open set contained in A and the closure of A, $\bar{A}$, is the smallest closed set containing A. If $X = \bar{A}$, we say A is dense in X.

Remark 6. (1) It's equivalent to define A° as the union of all open subset of A and $\bar{A}$ as the intersection of all closed supersets of A.

(2) The saying that $\mathbb{Q}$ is dense in $\mathbb{R}$ comes to us a lot, which is a good example for the definition of dense.

We list the following propositions.

Proposition 10.1.3. *Suppose X is a topology space and $A, B \subset X$*

(1) $A^\circ \subset A \subset \bar{A}$
(2) $(X \setminus A)^\circ = X \setminus \bar{A}$; $\quad \overline{(X \setminus B)} = X \setminus B^\circ$.
(3) $A^\circ \cup B^\circ \subset (A \cup B)^\circ$; $\quad A^\circ \cap B^\circ = (A \cap B)^\circ$
(4) *A is open* $\iff A = A^\circ$; *B is closed* $\iff B = \bar{B}$.
(5) $A \subset B \implies A^\circ \subset B^\circ, \bar{A} \subset \bar{B}$.
(6) $x \in A^\circ \iff \exists U_x \in \mathcal{U}(x) : U_x \subset A$.
(7) $x \in \bar{A} \iff \forall U_x \in \mathcal{U}(x) : U_x \cap A \neq \emptyset$.

To see how the decomposition works, we consider the following example.

Example 10.1.2. Consider the usual topology on $\mathbb{R}$ and let $I = [0, 1)$. Then we have:

$$\bar{I} = [0, 1], \quad I^\circ = (0, 1), \quad I' = [0, 1].$$

The boundary of I is defined as $\partial I = \bar{I} \setminus I^\circ = \{0, 1\}$. Notice that the derived set I' contains points that belong to I (namely $[0, 1)$) and points that do not (namely $\{1\}$). The boundary consists of two types of points: 1 is a limit point not in I, while 0 is a limit point in I but not an interior point. Then we have the following decompositions:

$$\bar{I} = I \cup I' = [0, 1) \cup [0, 1] = [0, 1],$$

and

$$\bar{I} = I^\circ \cup \partial I = (0, 1) \cup \{0, 1\} = [0, 1].$$

Thus I is neither open nor closed, and its closure can be expressed either as the union of I and its limit points, or as the union of its interior and its boundary.

Actually, this is not a special case*, we have the following results.

Definition 10.1.5 (Boundary)**.** Suppose X is a topology space and A is a subset of X. We define the boundary of A as $\partial A := \bar{A} \setminus A^\circ$.

Note 5. An intuitive understanding of the boundary is that it contains the points that are "close" to A but not necessarily in A. The meaning of "close" is that the point is almost in the set, which is clearer in the context of metric space. An equivalent definition of the boundary is the set of points such that for any neighborhood of the point, it intersects both A and $X \setminus A$.

Theorem 10.1.4 (Decomposition I of Closure)**.** *Suppose X is a topology space and A is a subset of X. We have $\bar{A} = A^\circ \cup \partial A$.*

Theorem 10.1.5 (Decomposition II of Closure)**.** *Suppose X is a topology space and A is a subset of X. We have $\bar{A} = A \cup A'$.*

Proof. We first show that $A \cup A' \subset \bar{A}$. It's trivial that $A \subset \bar{A}$, we only need to show

* You may also notice that $\bar{I} = I^\circ \cup I'$, but it's not always the case. A counterexample is $\{0\} \cup [1, 2]$, as you may verify.

$A' \subset \bar{A}$. For any $x \in A'$ and its neighborhood U_x, we have $U_x \cap (A \setminus \{x\}) \neq \emptyset$, which implies $U_x \cap A \neq \emptyset$. Since x is a limit point of A, we know that $x \in \bar{A}$.

We now show that $\bar{A} \subset A \cup A'$. For any $x \in \bar{A}$, if $x \in A$, then we are done. If $x \notin A$, then for any neighborhood U_x of x, we have $U_x \cap A \neq \emptyset$. Since $x \notin A$, we have $U_x \cap (A \setminus \{x\}) \neq \emptyset$, which implies that x is a limit point of A. Thus $x \in A'$ and we are done. □

We close this subsection by listing some important properties of the boundary.

10.1.3 Topological Basis

We've encountered the concept of basis before. The idea of basis is to find some elements to characterize elements with operations in the space. An example of this idea is the vector space, where we use the linear combination. Now, the operations concerning sets is union and intersection. We try to find a set of open sets such that we can construct all open set using union and intersection. This is the idea of basis in topology space.

Definition 10.1.6 (Basis)**.** Let X be a topology set and $\mathcal{B}$ is set of open sets. If $\forall U \in \tau : \exists \mathcal{U} \subset \mathcal{B}$ s.t. $U = \bigcup_{B \in \mathcal{U}} B$, then we say $\mathcal{B}$ is a topological basis and induces a topology τ on X:

$$\tau := \left\{ \bigcup_{B \in \mathcal{U}} B : \mathcal{U} \subset \mathcal{B} \right\} \tag{10.1}$$

One should verify that τ is indeed a topology.

We also have an equivalent definition.

Definition 10.1.7 (an Equivalent Definition of Topological Basis)**.** Let X be a topology set and $\mathcal{B}$ is set of nonempty subsets of X. We say $\mathcal{B}$ is a basis for the topology τ if the followings hold:

(1) $\forall x \in X : \exists B \in \mathcal{B}$ s.t. $x \in B$
(2) $\forall B_1, B_2 \in \mathcal{B} : \exists B \in \mathcal{B}$ s.t. $B \subset B_1 \cap B_2$

(2) has an equivalent form: $\forall B_1, B_2 \in \mathcal{B}, \forall x \in B_1 \cap B_2 : \exists B \in \mathcal{B}$ s.t. $x \in B \subset B_1 \cap B_2$.

Remark 7. To prove the equivalence, we only have to show the induced topology is the same with (10.1).

The proof is trivial as we only use definition. Here we consider what the 2 conditions mean. We consider the induced topology τ'. The first condition ensures that $X \in \tau'$. And the second one actually ensures that τ' is closed under finite intersection. Consider $U, V \in \tau'$, we have $U = \bigcup_{i=1}^{n_U} U_i$ and $V = \bigcup_{i=1}^{n_V} V_i$ where $U_i, V_i \in \mathcal{B}$. Then we have

$$U \cap V = \bigcup_{i=1}^{n_U} \bigcup_{j=1}^{n_V} (U_i \cap V_j)$$

Since $U_i \cap V_j \in \mathcal{B}$ by (2) again, we have $U \cap V \in \tau'$.

For a topology space, there may be many different basis. Take $\mathbb{R}^2$ as an example, the open balls and the open rectangles are both basis for the usual topology on $\mathbb{R}^2$. The following theorem shows which basis may induce the same topology.

Definition 10.1.8 (Equivalence of Basis)**.** Let X be a set and $\mathcal{B}_1, \mathcal{B}_2$ be two basis on X. We say they are equivalent if:

- $\forall B \in \mathcal{B}_1$ and $x \in B : \exists B' \in \mathcal{B}_2$ s.t. $x \in B' \subset B$.
- $\forall B \in \mathcal{B}_2$ and $x \in B : \exists B' \in \mathcal{B}_1$ s.t. $x \in B' \subset B$.

Theorem 10.1.6. *Let X be a set and $\mathcal{B}_1, \mathcal{B}_2$ be two basis on X. If $\mathcal{B}_1, \mathcal{B}_2$ are equivalent, they induce the same topology.*

Proof. Suppose the topology induced by $\mathcal{B}_1$ is τ_1 and the topology induced by $\mathcal{B}_2$ is τ_2. We try to show $\tau_1 = \tau_2$ by showing that $\tau_1 \subset \tau_2$ and $\tau_2 \subset \tau_1$.

We first show that $\tau_1 \subset \tau_2$. For any $U \in \tau_1$, by the definition of equivalent basis, we have $U = \bigcup_{x \in U} B_x$ where $B_x \in \mathcal{B}_2$. Thus $U \in \tau_2$. $\tau_2 \subset \tau_1$ can be shown similarly. $\square$

10.2 Continuous Function

10.2.1 Continuity in Metric Space

Definition 10.2.1 (Point-wise Continuity)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is continuous at a point $x_0 \in X$ if

$$\forall \varepsilon > 0, \exists \delta = \delta(x_0, \varepsilon) > 0 : d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

Using the language of neighborhood, we can give an equivalent definition of continuity.

Proposition 10.2.1. *Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is continuous at a point $x_0 \in X$ if for every neighborhood $V \subset Y$ of $f(x_0)$, there exists a neighborhood $U \subset X$ of x_0 such that $f(U) \subset V$. You may also use balls to characterize the neighborhood.*

If a function is continuous at every point of a subset A, we say the function is continuous on A.

Definition 10.2.2 (Lipschitz Continuity)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is Lipschitz continuous if there exists a constant $l > 0$ such that for any $x_1, x_2 \in X$, we have

$$d_Y(f(x_1), f(x_2)) \leq l d_X(x_1, x_2), \qquad x_1, x_2 \in X.$$

We call the smallest l the Lipschitz constant of f.

We'll see in the next proposition that Lipschitz continuity is a stronger condition than continuity.

Proposition 10.2.2. *Every Lipschitz continuous function is continuous.*

Proof. For any given x_0, let $\delta(\varepsilon) = \varepsilon/l$. Then for any x such that $d_X(x, x_0) < \delta(\varepsilon)$, we have

$$d_Y(f(x), f(x_0)) \leq l \cdot d_X(x, x_0) < l \cdot \delta(\varepsilon) = \varepsilon.$$

This shows that f is continuous at x_0. Since x_0 is arbitrary, f is continuous on X. □

Remark 8. Notice that in the proof, the choice of $\delta(\varepsilon)$ has nothing to do with x_0. This is stronger than continuity and we call it **uniform continuity**. Thus, every Lipschitz continuous function is uniformly continuous. This is a first touch on the concept of "uniformity", we'll see more about it in the following context.

Definition 10.2.3 (Uniform Continuity)**.** Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is uniformly continuous if

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0 : d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

10.2.2 Continuity

The above definitions rely on the metric. However, metric is not necessary and we turn to the definition of continuity in topology space. The definition is constructed similarly.

Definition 10.2.4 (Continuity)**.** Let (X, τ_X) and (Y, τ_Y) be topology spaces. A function $f : X \to Y$ is continuous at a point $x_0 \in X$ if for every neighborhood $V \subset Y$ of $f(x_0)$, there exists a neighborhood $U \subset X$ of x_0 such that $f(U) \subset V$. We say f is continuous on X if it's continuous at every point of X.

We list some characterizations of continuity.

Proposition 10.2.3. *Let (X, τ_X) and (Y, τ_Y) be topology spaces. The followings are equivalent:*

(1) f is continuous on X.
(2) For any open set $U \subset Y$, $f^{-1}(U)$ is open in X.
(3) For any closed set $V \subset Y$, $f^{-1}(V)$ is closed in X.
(4) For a basis $\mathcal{B}_Y$ of τ_Y and any $B \in \mathcal{B}_Y$, $f^{-1}(B)$ is open in X.

Proof. (1)⇒(2) For any open set $U \subset Y$ and $x \in f^{-1}(U)$, U is a neighborhood of $f(x)$. Then by definition, there exists a neighborhood $V \in \mathcal{U}(x)$ such that $f(V) \subset U$, implying that x is an interior point. Since x is arbitrary, $f^{-1}(U)$ is open in X.

(2)⇒(3) First, notice that $f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$ for any $V \subset Y$. If V is closed in Y, then $Y \setminus V$ is open in Y. By (2), we have that $f^{-1}(Y \setminus V)$ is open in X. Therefore, $f^{-1}(V) = X \setminus f^{-1}(Y \setminus V)$ is closed in X.

(3)⇒(4) Suppose $B \in \mathcal{B}_Y$, then $Y \setminus B$ is closed in Y. By (3), we have that $f^{-1}(Y \setminus B)$ is closed in X. Therefore, $f^{-1}(B) = X \setminus f^{-1}(Y \setminus B)$ is open in X.

(4)⇒(1) Only have to notice that any open set can be written as the union of basis elements and the preimage of a union is the union of preimages. □

Now we list some important properties of continious function.

Theorem 10.2.4. *Suppose $f : X \to Y$ is continuous and $A \subset X$, then*

$$x \in \bar{A} \implies f(x) \in \overline{f(A)}.$$

Proof. We prove by contradiiction. □

Theorem 10.2.5 (Sequential Continuity)**.** *$f : X \to Y$ is continuous $\implies f(x_n) \to f(x_0)$ when $x_n \to x_0$.*

Remark 9. The reverse proposition also holds, i.e., if for any sequence in X converging to x_0, the sequence $\{f(x_n)\}$ converges to $f(x_0)$, then f is continuous.

Proof. □

10.2.3 Homomorphism

10.3 Compactness

10.3.1 the Basics

10.3.2 Compactness of Product Space

10.3.3 Compactness of Induced Topology Space

10.4 Connectedness

10.5 Constructing Topology

10.5.1 Topology Subspace

Similar to other spaces, we can define its subspace.

Definition 10.5.1 (Topology Subspace)**.** Suppose $Y \subset X$, let $\tau_Y := \{U \cap Y : U \in \tau\}$, then (Y, τ_Y) is a topology space. We say a set is open in the subspace if it is an element of τ_Y and a set is closed in the subspace if its complement in Y is open in the subspace.

10.5.2 Product Space

10.5.3 Quotient Space

10.5.4 Topology Space Induced by Metric

Part IV

Differentiation in One Variable

Part V

More in Convergence: Uniform Convergence

Appendix A

Notation

Appendix B

Supplementary Scripts

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